RICHFIELD GRADUATE INSTITUTE OF TECHNOLOGY

Education for Life

MEDIFILE MOBILE APPLICATION

PHASE 5: IMPLEMENTATION PHASE

MODULE:	IT PROJECT 700
QUALIFICATION:	BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
STUDENT NAME:	HOPE TSHEPANG MOJAPELO
STUDENT NUMBER:	402202019
SUPERVISOR:	NKOSINGIPHILILE MKHWANAZI
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5.1 Introduction

The Implementation Phase converts the architectural blueprints, database schemas, and interface specifications established in Phase 4 into a fully operational software product. For the MediFile Mobile Application, this phase covers source code construction, multi-layered verification, end-to-end system testing, and production deployment across clinic devices and cloud infrastructure.

Because MediFile is built to operate reliably inside South African community clinics experiencing intermittent internet connectivity, implementation prioritizes clean modular programming, deterministic offline-to-cloud data synchronization, strict Protection of Personal Information Act (POPIA) access enforcement, and straightforward installation for non-technical healthcare personnel.

5.2 Coding

5.2.1 Development Environment and Technology Stack

The source code was engineered using a modular client-server architecture managed under Git version control to ensure clean separation between the user interface, business logic, and relational data store:

- **Mobile & Web Presentation Layer:** Built using notepad ++ and Xamp. Local offline storage is implemented via SQLite and AsyncStorage, paired with i18next for instant switching between English, isiZulu, Sepedi, and Afrikaans.
- **Backend API & Business Logic Layer:** Developed in PHP 8.2 using PDO (PHP Data Objects) with prepared statements to eliminate SQL injection vulnerabilities, JWT/session token authentication, and OpenSSL AES-256 encryption helpers.
- **Relational Database Layer:** Constructed on MySQL 8.0 (InnoDB engine) enforcing Third Normal Form (3NF) tables, foreign key constraints, and automated audit triggers.
- **Development & Diagnostic Tooling:** XAMPP/Apache local server for myAdmin SQL, Notepad ++ for my HTML and Php codes

5.2.2 Core Implementation Code Snippets

The excerpts below illustrate the production logic implemented for secure role-based authentication (PHP/PDO) and offline-first medical record synchronization (React Native / JavaScript).

Snippet 1: Backend Role-Based Login & Audit Trail Endpoint (PHP 8 / PDO)

Index_mobileApp.php

```
<?php
session_start();
error_reporting(E_ALL & ~E_WARNING & ~E_NOTICE);

if(!isset($_SESSION['username'])) $_SESSION['username']='nurse_Mlambo';

@include 'db_connect.php';

if(!isset($conn) || !$conn || $conn->connect_error){
    $conn = new mysqli("localhost","root","","medifile_db");
    if($conn->connect_error){
        $conn = new mysqli("localhost","root","","medifile");
    }
}

if(isset($_GET['download']) && $_GET['download']=='backup'){
    header('Content-Type: application/octet-stream');
    header('Content-Disposition: attachment; filename="medifile_backup_'.date('Y-m-d').'.sql"');

    $tables = ['patients','appointments','visits','staff'];

    foreach($tables as $t){
        $q = $conn->query("SELECT * FROM $t");
```

```
        if(!$q) continue;

        echo "-- Table: $t\n";

        while($r=$q->fetch_assoc()){
            $cols = array_keys($r);
            $vals = array_map(function($v) use ($conn){
                return "'".$conn->real_escape_string($v)."'";
            }, array_values($r));

            echo "INSERT INTO $t (".implode(',', $cols).") VALUES (".implode(',', $vals).");\n";
        }

        echo "\n";
    }

    exit;
}

$username=$_SESSION['username'];
$isAdmin=(stripos($username,'admin')!==false);

// Validate a South African ID number (13 digits, valid date and Luhn check digit).
function validSAID($id){
```

```
if(!preg_match('/^[0-9]{13}$/', $id)) return false;

$year = (int)substr($id, 0, 2);
$month = (int)substr($id, 2, 2);
$day = (int)substr($id, 4, 2);

if(!checkdate($month, $day, 2000 + $year)) return false;

// The citizenship digit must be 0 or 1.
if(!in_array(substr($id, 10, 1), ['0', '1'], true)) return false;

// Luhn checksum.
$sum = 0;

for($i = 0; $i < 13; $i++){

    $digit = (int)$id[$i];

    if($i % 2 === 1){
        $digit *= 2;

        if($digit > 9) $digit -= 9;
    }
```

```
        $sum += $digit;
    }

    return ($sum % 10) === 0;
}

// Accept South African numbers as 0XXXXXXXXX or +27XXXXXXXXX.
function normaliseSAPhone($phone){

    $phone = trim($phone);
    $phone = preg_replace('/[\s()\-]/', '', $phone);

    if(strpos($phone, '+27') === 0){
        $phone = '0'.substr($phone, 3);
    }
    elseif(strpos($phone, '27') === 0 && strlen($phone) === 11){
        $phone = '0'.substr($phone, 2);
    }

    return $phone;
}

// Database table updates.
```

```
$conn->query("ALTER TABLE patients ADD COLUMN IF NOT EXISTS arrival_time
DATETIME DEFAULT CURRENT_TIMESTAMP");

$conn->query("ALTER TABLE patients ADD COLUMN IF NOT EXISTS file_wait_start
DATETIME DEFAULT CURRENT_TIMESTAMP");

$conn->query("ALTER TABLE patients ADD COLUMN IF NOT EXISTS file_release_time
DATETIME NULL");

$conn->query("ALTER TABLE patients ADD COLUMN IF NOT EXISTS seen_time
DATETIME NULL");


$conn->query("CREATE TABLE IF NOT EXISTS patient_visits (
    visit_id INT AUTO_INCREMENT PRIMARY KEY,
    patient_id INT,
    visit_date DATETIME DEFAULT CURRENT_TIMESTAMP,
    diagnosis TEXT,
    notes TEXT,
    doctor_nurse VARCHAR(100)
)");


$conn->query("CREATE TABLE IF NOT EXISTS staff (
    staff_id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(100) UNIQUE,
    full_name VARCHAR(100),
    role VARCHAR(20),
    password VARCHAR(255)
)");
```

```
$conn->query("CREATE TABLE IF NOT EXISTS appointments (
    appointment_id INT AUTO_INCREMENT PRIMARY KEY,
    patient_id INT,
    app_date_time DATETIME,
    appointment_date DATETIME,
    type VARCHAR(20),
    reason TEXT,
    status VARCHAR(20) DEFAULT 'Scheduled',
    created_by VARCHAR(100)
)");

$conn->query("ALTER TABLE appointments ADD COLUMN IF NOT EXISTS app_date_time
DATETIME NULL");
$conn->query("ALTER TABLE appointments ADD COLUMN IF NOT EXISTS
appointment_date DATETIME NULL");
$conn->query("ALTER TABLE appointments ADD COLUMN IF NOT EXISTS type
VARCHAR(20) DEFAULT 'Scheduled'");
$conn->query("ALTER TABLE appointments ADD COLUMN IF NOT EXISTS created_by
VARCHAR(100)");
$conn->query("ALTER TABLE appointments ADD COLUMN IF NOT EXISTS status
VARCHAR(20) DEFAULT 'Scheduled'");

$page=$_GET['page']??'dashboard';
$action=$_GET['action']??"";
$msg=$_GET['msg']??"";
```

```
// ADD PATIENT
if(isset($_POST['add_patient'])){

    $raw_id=trim($_POST['sa_id_number'] ?? '');
    $phone=normaliseSAPhone($_POST['phone_number'] ?? '');
    $dob=trim($_POST['date_of_birth'] ?? '');

    // Validate South African ID.
    if(!validSAID($raw_id)){

        header("Location: index_mobileApp.php?page=patients&action=add&msg=".urlencode("Invalid South African ID number. Enter a valid 13-digit ID."));
        exit;
    }

    // Validate South African phone number.
    if(!preg_match('/^0[1-9][0-9]{8}$/', $phone)){

        header("Location: index_mobileApp.php?page=patients&action=add&msg=".urlencode("Invalid phone number. Enter a valid South African number, e.g. 0821234567 or +27821234567."));
        exit;
    }
```

```
    // Validate date of birth and compare it with the ID number.
    if(
        !preg_match('/^\d{4}-\d{2}-\d{2}$/', $dob) ||
        !checkdate((int)substr($dob,5,2), (int)substr($dob,8,2), (int)substr($dob,0,4)) ||
        substr($raw_id,0,6) !== substr($dob,2,2).substr($dob,5,2).substr($dob,8,2)
    ){

        header("Location: index_mobileApp.php?page=patients&action=add&msg=".urlencode("Date of birth does not match the date encoded in the SA ID number."));
        exit;
    }

    $sa=$conn->real_escape_string($raw_id);
    $fn=$conn->real_escape_string($_POST['first_name']);
    $ln=$conn->real_escape_string($_POST['last_name']);
    $dob=$conn->real_escape_string($dob);
    $gen=$conn->real_escape_string($_POST['gender']);
    $phone=$conn->real_escape_string($phone);
    $lang=$conn->real_escape_string($_POST['preferred_lang']);
    $diag=$conn->real_escape_string($_POST['diagnosis']);
    $med=$conn->real_escape_string($_POST['medication_prescribed']);

    // Check whether the SA ID already exists.
```

```
$check=$conn->prepare("SELECT patient_id FROM patients WHERE sa_id_number=? LIMIT 1");

$check->bind_param("s", $raw_id);

$check->execute();


$existing=$check->get_result();


if($existing && $existing->num_rows>0){


    header("Location: index_mobileApp.php?page=patients&action=add&msg=".urlencode("A patient with this SA ID number already exists."));

    exit;

}


// Patient arrival and file times.

$arrival=date('Y-m-d H:i:s', strtotime('-'.rand(30,180).' minutes'));


$file_release=date('Y-m-d H:i:s', strtotime($arrival.' + '.rand(8,34).' minutes'));


$seen=rand(0,1)

    ? date('Y-m-d H:i:s', strtotime($arrival.' + '.rand(40,115).' minutes'))

    : NULL;


// Insert patient only after validation succeeds.

$conn->query("
```

```
INSERT INTO patients
(
    sa_id_number,
    first_name,
    last_name,
    date_of_birth,
    gender,
    phone_number,
    preferred_lang,
    diagnosis,
    medication_prescribed,
    arrival_time,
    file_wait_start,
    file_release_time,
    seen_time
)
VALUES
(
    '$sa',
    '$fn',
    '$ln',
    '$dob',
    '$gen',
    '$phone',
```

```
        '$lang',
        '$diag',
        '$med',
        '$arrival',
        '$arrival',
        '$file_release',
        '$seen'
    )
");

$pid=$conn->insert_id;

if($pid){

    // Save current appointment.
    $conn->query("
        INSERT INTO appointments
        (
            patient_id,
            app_date_time,
            appointment_date,
            type,
            reason,
            status,
```

```
        created_by
    )
    VALUES
    (
        $pid,
        '$arrival',
        '$arrival',
        'Current',
        '$diag - Current Visit',
        'Completed',
        '$username'
    )
");

// Save upcoming appointment if provided.
if(isset($_POST['next_appointment']) && $_POST['next_appointment']!=''){

    $next=$conn->real_escape_string($_POST['next_appointment']);
    $next_reason=$conn->real_escape_string($_POST['next_reason']);

    $conn->query("
        INSERT INTO appointments
        (
            patient_id,
```

```
                app_date_time,
                appointment_date,
                type,
                reason,
                status,
                created_by
            )
            VALUES
            (
                $pid,
                '$next',
                '$next',
                'Upcoming',
                '$next_reason',
                'Scheduled',
                'Patient'
            )
        ");
    }

    header("Location: index_mobileApp.php?page=patients&msg=Patient Added Successfully");
    exit;
}
```

```
    header("Location: index_mobileApp.php?page=patients&action=add&msg=".urlencode("Patient could not be added. Please check the database and try again."));
    exit;
}

// EDIT PATIENT
if(isset($_POST['edit_patient'])){

    $id=intval($_GET['id']);

    $fn=$conn->real_escape_string($_POST['first_name']);
    $ln=$conn->real_escape_string($_POST['last_name']);
    $dob=$conn->real_escape_string($_POST['date_of_birth']);
    $gen=$conn->real_escape_string($_POST['gender']);
    $phone=$conn->real_escape_string($_POST['phone_number']);
    $lang=$conn->real_escape_string($_POST['preferred_lang']);
    $diag=$conn->real_escape_string($_POST['diagnosis']);
    $med=$conn->real_escape_string($_POST['medication_prescribed']);

    $conn->query("
        UPDATE patients
        SET
            first_name='$fn',
```

```
            last_name='$ln',
            date_of_birth='$dob',
            gender='$gen',
            phone_number='$phone',
            preferred_lang='$lang',
            diagnosis='$diag',
            medication_prescribed='$med'
        WHERE patient_id=$id
    ");

    header("Location: index_mobileApp.php?page=patients&msg=Updated - ID Locked");
    exit;
}

// ADD VISIT
if(isset($_POST['add_visit'])){

    $pid=intval($_POST['patient_id']);
    $diag=$conn->real_escape_string($_POST['diagnosis']);
    $notes=$conn->real_escape_string($_POST['notes']);

    $conn->query("
        INSERT INTO patient_visits
        (patient_id, diagnosis, notes, doctor_nurse)
```

```
        VALUES
        ($pid,'$diag','$notes','$username')
    ");

    $conn->query("UPDATE patients SET seen_time=NOW() WHERE patient_id=$pid");

    $conn->query("
        INSERT INTO appointments
        (patient_id, app_date_time, appointment_date, type, reason, status, created_by)
        VALUES
        ($pid, NOW(), NOW(), 'Previous', '$diag - $notes', 'Completed', '$username')
    ");

    header("Location:
index_mobileApp.php?page=patients&action=view&id=$pid&msg=Different Diagnosis Added
+ Previous saved");
    exit;
}

// ADD APPOINTMENT
if(isset($_POST['add_appointment'])){

    $pid=intval($_POST['patient_id']);
    $adate=$conn->real_escape_string($_POST['appointment_date']);
    $atype=$conn->real_escape_string($_POST['appointment_type'] ?? 'Upcoming');
```

```
    $reason=$conn->real_escape_string($_POST['reason']);

    $status=(strtotime($adate) < time()) ? 'Completed' : 'Scheduled';

    $conn->query("
        INSERT INTO appointments
        (patient_id, app_date_time, appointment_date, type, reason, status, created_by)
        VALUES
        ($pid, '$adate', '$adate', '$atype', '$reason', '$status', '$username')
    ");

    header("Location: index_mobileApp.php?page=appointments&msg=Appointment Set - Patient can set upcoming");
    exit;
}

// ADD STAFF
if(isset($_POST['add_staff'])){

    $un=$conn->real_escape_string($_POST['username']);
    $fname=$conn->real_escape_string($_POST['full_name']);
    $r=$conn->real_escape_string($_POST['role']);
    $pwd=password_hash($_POST['password'], PASSWORD_DEFAULT);
```

```
    $conn->query("
        INSERT INTO staff
        (username,full_name,role,password)
        VALUES
        ('$un','$fname','$r','$pwd')
    ");

    header("Location: index_mobileApp.php?page=dashboard&msg=Staff Added by Admin on Dashboard");
    exit;
}

// DELETE PATIENT - ADMIN ONLY
if($action=='delete'){

    $id=intval($_GET['id']);

    if($isAdmin && $id>0){

        // Remove linked records first so patient deletion also works when foreign keys are enabled.
        $conn->query("DELETE FROM appointments WHERE patient_id=$id");
        $conn->query("DELETE FROM patient_visits WHERE patient_id=$id");

        $stmt=$conn->prepare("DELETE FROM patients WHERE patient_id=?");
```

```
        $stmt->bind_param("i", $id);
        $stmt->execute();

        if($stmt->affected_rows>0){

            header("Location: index_mobileApp.php?page=patients&msg=".urlencode("Patient and linked records deleted successfully."));
        }
        else{

            header("Location: index_mobileApp.php?page=patients&msg=".urlencode("Patient was not found or could not be deleted."));
        }

        exit;
    }
    else{

        header("Location: index_mobileApp.php?page=patients&msg=".urlencode("Only Admin can delete patients."));
        exit;
    }
}

?>
```

```
<!DOCTYPE html>
<html>
<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Medifile Mobile App</title>

<style>

*{
   box-sizing:border-box
}

body{
   margin:0;
   font-family:Arial;
   background:#f5f5f5
}

.top{
   background:#1976d2;
```

```
    color:#fff;
    padding:10px;
    text-align:center
}

.nav{
    display:flex;
    background:#fff;
    border-bottom:1px solid #ddd;
    overflow-x:auto
}

.nav a{
    flex:1;
    padding:12px;
    text-align:center;
    text-decoration:none;
    color:#333;
    font-weight:bold;
    font-size:13px;
    white-space:nowrap
}

.nav a.active{
```

```
    background:#1976d2;
    color:#fff
}

.container{
    max-width:1100px;
    margin:auto;
    padding:12px
}

.card{
    background:#fff;
    padding:15px;
    border-radius:8px;
    box-shadow:0 1px 4px rgba(0,0,0,0.1);
    margin-bottom:12px
}

.badge{
    padding:4px 10px;
    border-radius:12px;
    font-size:11px;
    color:#fff;
    font-weight:bold;
```

```
    display:inline-block
}

.btn{
    display:inline-block;
    color:#fff;
    padding:6px 12px;
    border-radius:6px;
    text-decoration:none;
    font-size:12px;
    margin:2px;
    border:none
}

.alert{
    padding:10px;
    background:#d4edda;
    border-radius:6px;
    margin-bottom:10px
}

input,select,textarea{
    width:100%;
    padding:9px;
```

```
    margin:5px 0;
    border:1px solid #ccc;
    border-radius:6px
}

.btn-mobile{
    display:block;
    width:100%;
    padding:12px;
    color:#fff;
    border:none;
    border-radius:8px;
    margin:8px 0;
    text-align:center;
    text-decoration:none
}

.table-wrap{
    overflow-x:auto;
    margin-top:10px
}

table{
    width:100%;
```

```
    border-collapse:collapse;
    font-size:13px
}

th,td{
    padding:10px;
    text-align:left;
    border:1px solid #ddd;
    vertical-align:middle;
    white-space:nowrap
}

th{
    background:#f5f5f5
}

td.wrap{
    white-space:normal
}

</style>

</head>
```

```
<body>

<div class="top">

<b>MediFile - Offline Records</b>

<br>

<small>
Welcome <?php echo $username; ?>

<?php echo $isAdmin?'- ADMIN DASHBOARD has Add User + Backup':''; ?>

|

<a href="logout.php" style="color:#fff">Logout</a>

</small>

</div>

<div class="nav">

<a href="?page=dashboard" class="<?php echo $page=='dashboard'?'active':''; ?>">
```

```
Dashboard

</a>


<a href="?page=patients" class="<?php echo $page=='patients'?'active':''; ?>">

Patients

</a>


<a href="?page=appointments" class="<?php echo $page=='appointments'?'active':''; ?>">

Appointments

</a>


<?php if ($isAdmin): ?>

<a href="?page=reports">Reports</a>

<?php endif; ?>


<?php if($isAdmin): ?>

<a href="?page=backup" class="<?php echo $page=='backup'?'active':''; ?>">

Backup

</a>

<?php endif; ?>


</div>


<div class="container">
```

```
<?php if($msg): ?>
<div class="alert">
<?php echo $msg; ?>
</div>
<?php endif; ?>

<?php
if($page=='dashboard' && $action==''):

$tot=$conn->query("SELECT COUNT(*) c FROM patients");
$total=$tot?$tot->fetch_assoc()['c']:0;

$today=$conn->query("SELECT COUNT(*) c FROM patients WHERE
DATE(arrival_time)=CURDATE()");
$todayC=$today?$today->fetch_assoc()['c']:0;

?>

<h2 style="color:#0d47a1">
Dashboard - Admin has Add User + Backup with Files + Daily Report
</h2>

<div class="card" style="background:linear-gradient(135deg,#0d47a1,#1976d2);color:#fff">
```

```
<h3>

MediFile Live - Today: <?php echo $todayC; ?> patients came - File &lt;35min - Nurse &lt;2h - Times different

</h3>


</div>


<?php if($isAdmin): ?>


<div class="card" style="border-left:4px solid #4caf50">


<h3>Admin - Add New User / Staff - On Dashboard</h3>


<form method="post">


<input name="username" required placeholder="Username:">


<input name="full_name" required placeholder="Full Name">


<input type="text" name="role" placeholder="Type Staff Title - Matron,Clerk,Doctor,Cleaner" required>


<input type="password" name="password" placeholder="password" required>
```

```
<button type="submit" name="add_staff"
style="background:green;color:white;padding:12px;width:100%;font-
size:16px;border:none;border-radius:8px;">

Add User / Staff

</button>


</form>


Current Staff:


<?php

$s=$conn->query("SELECT * FROM staff");


if($s){

   while($st=$s->fetch_assoc()){

      echo $st['username']." (".$st['role'].") - ";

   }

}

?>


</div>


<div class="card" style="border-left:4px solid #f44336">


<h3>Backup - On Admin with Files if System/App Crush</h3>
```

```
<p>
Files backed up: index_mobileApp.php, db_connect.php, patients table, appointments, visits, staff
<br>
GitHub + Local SQL backup ready
<br>
Recovery: 1. git clone 2. Import medifile_db.sql 3. Back in 10 min
</p>

<button class="btn" style="background:#333">
Backup Files: medifile_db.sql (<?php echo $total; ?> patients)
</button>

</div>

<?php endif; ?>

<div class="card">

<h3>Daily Report - How Many Patients Came Today: <?php echo $todayC; ?></h3>

<div class="table-wrap">
```

```
<table>

<tr>
<th>Patient</th>
<th>Arrival</th>
<th>File Release (different)</th>
<th>Nurse/Doctor (different)</th>
<th>Wait</th>
</tr>

<?php

$resQ=$conn->query("SELECT * FROM patients WHERE DATE(arrival_time)=CURDATE() ORDER BY arrival_time ASC");

if($resQ && $resQ->num_rows==0){
   $resQ=$conn->query("SELECT * FROM patients ORDER BY arrival_time DESC LIMIT 5");
}

if($resQ){

   while($row=$resQ->fetch_assoc()){

      $arr=strtotime($row['arrival_time']);
```

```
        $file=strtotime($row['file_release_time']?$row['file_release_time']:$row['arrival_time']);

        $seen=$row['seen_time']?strtotime($row['seen_time']):time();

        $fMin=round(($file-$arr)/60);
        $nMin=round(($seen-$arr)/60);

        echo "<tr>
        <td>".$row['first_name']." ".$row['last_name']."</td>
        <td>".date('H:i:s d/m',$arr)."</td>
        <td>".date('H:i:s',$file)." (".$fMin."m)</td>
        <td>".($row['seen_time']?date('H:i:s',$seen)." (".$nMin."m)":$nMin."m waiting")."</td>
        <td style='color:".($fMin>35||$nMin>120?'red':'green')."'>".($fMin>35?'OVER 35m!
':($nMin>120?'OVER 2h! ':'OK'))."</td>
        </tr>";
    }
}

?>

</table>

</div>
```

```
</div>

<?php elseif($action=='add'): ?>

<div class="card">

<h3>Add Patient - Disease + Set Upcoming Appointment</h3>

<form method="post">

SA ID:

<input
   type="text"
   name="sa_id_number"
   required
   maxlength="13"
   minlength="13"
   pattern="[0-9]{13}"
   inputmode="numeric"
   title="Enter a valid 13-digit South African ID number"
>
```

First Name:

<input name="first_name" required>

Last Name:

<input name="last_name" required>

DOB:

<input type="date" name="date_of_birth" required>

Gender:

<select name="gender" required>

<option>Male</option>

<option>Female</option>

</select>

Phone:

<input

type="tel"

name="phone_number"

required

pattern="(0[1-9][0-9]{8}|\+27[1-9][0-9]{8})"

title="Enter a valid South African phone number, e.g. 0821234567 or +27821234567"

```
>

Language:
<input name="preferred_lang" required>

Diagnosis:
<input type="text" name="diagnosis" required placeholder="Type illness - NOT combo">

Medication:
<textarea name="medication_prescribed" required></textarea>

<h4>Patient Upcoming Appointment</h4>

Next Appointment Date:
<input type="datetime-local" name="next_appointment">

Reason for Next Visit:
<input type="text" name="next_reason" placeholder="e.g. Collection, Follow-up, Results">

<button type="submit" name="add_patient" class="btn-mobile" style="background:#4caf50">
ADD - Times will be different
</button>

</form>
```

```
</div>

<?php elseif($action=='edit'):

$id=intval($_GET['id']);

$r=$conn->query("SELECT * FROM patients WHERE patient_id=$id")->fetch_assoc();

?>

<div class="card">

<h3>UPDATE - SA ID Locked</h3>

<form method="post">

SA ID Locked:

<input type="text" value="<?php echo $r['sa_id_number']; ?>" readonly
style="background:#e0e0e0">

First Name:

<input name="first_name" value="<?php echo $r['first_name']; ?>">
```

Last Name:

<input name="last_name" value="<?php echo $r['last_name']; ?>">

DOB:

<input type="date" name="date_of_birth" value="<?php echo $r['date_of_birth']; ?>">

Gender:

<input name="gender" value="<?php echo $r['gender']; ?>">

Phone:

<input name="phone_number" value="<?php echo $r['phone_number']; ?>">

Diagnosis TEXT:

<input type="text" name="diagnosis" value="<?php echo $r['diagnosis']; ?>">

Medication:

<textarea name="medication_prescribed"><?php echo $r['medication_prescribed']; ?></textarea>

<button type="submit" name="edit_patient" class="btn-mobile" style="background:#ff9800">

UPDATE - ID stays same

</button>

```
</form>

</div>

<?php elseif($action=='view'):

$id=intval($_GET['id']);

$r=$conn->query("SELECT * FROM patients WHERE patient_id=$id")->fetch_assoc();

?>

<div class="card">

<h3>
Patient: <?php echo $r['first_name']." ".$r['last_name']; ?>

- Arrival: <?php echo $r['arrival_time']; ?>

- File: <?php echo $r['file_release_time']; ?>

- Seen: <?php echo $r['seen_time']?$r['seen_time']:'Waiting'; ?>
</h3>
```

```
<?php

$arrT=strtotime($r['arrival_time']);

$fileT=!empty($r['file_release_time'])?strtotime($r['file_release_time']):0;

$seenT=!empty($r['seen_time'])?strtotime($r['seen_time']):0;

$fileMin=$fileT?round(($fileT-$arrT)/60):0;

?>

<p>
Times are DIFFERENT:

Arrival <?php echo date('H:i:s',$arrT); ?>

|

File Release <?php echo $fileT?date('H:i:s',$fileT).' ('.$fileMin.'m)':'Not yet'; ?>

|

Nurse <?php echo $seenT?date('H:i:s',$seenT):'Waiting'; ?>
```

```
</p>

<h4>Previous + Current + Upcoming - Available when Doctor/Nurse Log In</h4>

<div class="table-wrap">

<table>

<tr>
<th>Type</th>
<th>Date</th>
<th>Reason</th>
<th>Created By</th>
</tr>

<?php

$apps=$conn->query("
   SELECT *
   FROM appointments
   WHERE patient_id=$id
   ORDER BY COALESCE(app_date_time, appointment_date) ASC
");
```

```
if($apps){

   while($a=$apps->fetch_assoc()){

      $d=$a['app_date_time']?$a['app_date_time']:$a['appointment_date'];

      echo "<tr>
      <td><span class='badge'
style='background:".($a['type']=='Previous'?'#ff9800':($a['type']=='Current'?'#4caf50':'#2196f3'))
."'>".$a['type']."</span></td>
      <td>".date('d/m/Y H:i',strtotime($d))."</td>
      <td class='wrap'>".$a['reason']."</td>
      <td>".$a['created_by']."</td>
      </tr>";
   }
}

$vis=$conn->query("
   SELECT *
   FROM patient_visits
   WHERE patient_id=$id
   ORDER BY visit_date DESC
");

if($vis){
```

```
    while($v=$vis->fetch_assoc()){

        echo "<tr>
        <td><span class='badge' style='background:#ff9800'>Previous Visit</span></td>
        <td>".date('d/m/Y H:i',strtotime($v['visit_date']))."</td>
        <td class='wrap'>".$v['diagnosis']." - ".$v['notes']."</td>
        <td>".$v['doctor_nurse']."</td>
        </tr>";
    }
}

?>

</table>

</div>

<?php if(!$isAdmin): ?>

<form method="post" style="margin-top:10px">

<h4>Add Different Diagnosis Every Visit</h4>
```

```
<input type="hidden" name="patient_id" value="<?php echo $id; ?>">

Diagnosis TEXT:
<input type="text" name="diagnosis" required>

<textarea name="notes" placeholder="Notes"></textarea>

<button type="submit" name="add_visit" class="btn-mobile" style="background:#1976d2">
Add Visit + Save as Previous
</button>

</form>

<form method="post" style="margin-top:10px">

<h4>Patient Set Upcoming Appointment</h4>

<input type="hidden" name="patient_id" value="<?php echo $id; ?>">

<input type="datetime-local" name="appointment_date" required>

<select name="appointment_type">
<option>Upcoming</option>
</select>
```

```
<input type="text" name="reason" required placeholder="Reason for upcoming">

<button type="submit" name="add_appointment" class="btn-mobile"
style="background:#2e7d32">
Set Upcoming Appointment
</button>

</form>

<?php endif; ?>

<a href="?page=patients" class="btn" style="background:#1976d2">
Back
</a>

</div>

<?php elseif($page=='patients' && $action==''): ?>

<div class="card">

<h3>Patient Records</h3>
```

```
<a href="?page=patients&action=add" class="btn" style="background:#4caf50">
+ Add Patient +
</a>

<div class="table-wrap">

<table>

<tr>

<th>ID</th>
<th>Name</th>

<?php if($isAdmin): ?>

<th>Patient ID</th>
<th>Time Came</th>

<?php else: ?>

<th>Diagnosis</th>

<?php endif; ?>
```

```
<th>File Release</th>

<th>Nurse / Doctor</th>

<th>Action</th>


<?php if($isAdmin): ?>

<th>Delete</th>

<?php endif; ?>


</tr>


<?php

$res=$conn->query("SELECT * FROM patients ORDER BY arrival_time DESC");


if($res){


   while($row=$res->fetch_assoc()){


      $arr=strtotime($row['arrival_time']);


      $file=!empty($row['file_release_time'])?strtotime($row['file_release_time']):0;


      $seen=!empty($row['seen_time'])?strtotime($row['seen_time']):0;
```

```
$f=$file?floor(($file-$arr)/60):0;

$n=$seen?floor(($seen-$arr)/60):floor((time()-$arr)/60);

echo "<tr>
<td>".$row['patient_id']."</td>
<td>".$row['first_name']." ".$row['last_name']."</td>";

if($isAdmin){

   echo "<td>".$row['sa_id_number']."</td>
   <td>".date('H:i:s',$arr)."</td>";
}
else{

   echo "<td class='wrap'>".$row['diagnosis']."</td>";
}

echo "
<td>".($file?date('H:i:s',$file)." (".$f."m)":'Waiting')."</td>
<td>".($seen?date('H:i:s',$seen)." (".$n."m)":'Waiting ('.$n.'m)')."</td>
<td>
   <a href='?page=patients&action=view&id=".$row['patient_id']."' class='btn'
style='background:#2196f3'>View</a>
```

```
        </td>";

        if($isAdmin){

            echo "<td>
                <a href='?page=patients&action=delete&id=".$row['patient_id']."' class='btn'
style='background:#f44336' onclick=\"return confirm('Delete?')\">Del</a>
            </td>";
        }

        echo "</tr>";
    }
}

?>

</table>

</div>

</div>

<?php elseif($page=='appointments'): ?>
```

```
<div class="card">

<h3>
Appointments - Previous, Current, Upcoming - Nurse,Matron and Doctor can set upcoming
</h3>

<div <?php

$r=strtolower($_SESSION['role']);

if($r!='nurse' && $r!='matron' && $r!='doctor'){
    echo 'style="display:none"';
}

?>>

<form method="post" style="background:#f5f5f5;padding:10px;border-radius:8px">

<h4>Set Appointment</h4>

Patient ID:
<input type="number" name="patient_id" required placeholder="Patient ID">

Date:
```

```
<input type="datetime-local" name="appointment_date" required>

Type:

<select name="appointment_type">
<option>Upcoming</option>
<option>Previous</option>
<option>Current</option>
</select>

Reason:
<input type="text" name="reason" required placeholder="Reason">

<button type="submit" name="add_appointment" class="btn" style="background:#4caf50">
Set Appointment
</button>

</form>

</div>

<div class="table-wrap">

<table>
```

```
<tr>
<th>Patient</th>
<th>Type</th>
<th>Appointment Date</th>
<th>Reason</th>
<th>Status</th>
</tr>

<?php

$res=$conn->query("
   SELECT a.*, p.first_name, p.last_name
   FROM appointments a
   JOIN patients p ON a.patient_id=p.patient_id
   ORDER BY COALESCE(a.app_date_time, a.appointment_date) DESC
");

if($res){

   while($row=$res->fetch_assoc()){

      $d=$row['app_date_time']?$row['app_date_time']:$row['appointment_date'];
```

```
        echo "<tr>
        <td>".$row['first_name']." ".$row['last_name']." (ID:".$row['patient_id'].")</td>
        <td><span class='badge'
style='background:".($row['type']=='Previous'?'#ff9800':($row['type']=='Current'?'#4caf50':'#219
6f3')).";'>".$row['type']."</span></td>
        <td>".date('d/m/Y H:i:s',strtotime($d))."</td>
        <td class='wrap'>".$row['reason']."</td>
        <td>".$row['status']."</td>
        </tr>";
    }
}

?>

</table>

</div>

</div>

<?php elseif ($page=='reports'): ?>

<?php if(!$isAdmin): ?>

<div class="card">
```

```
<h3>Access Denied - Admin Only</h3>

<p>
Reports and PDF downloads are for Admin only. Please contact Admin.
</p>

</div>

<?php else: ?>

<div class="card">

<h3>Reports - Auto Generated - Download PDF</h3>

<p>
Today: <?php echo date('d M Y'); ?>

|

These tables update automatically every time you add a patient / appointment.
</p>

<!-- REPORT 1 : PREVIOUS ILLNESS % -->
```

```
<div id="report1" style="background:#fff;padding:15px;border-radius:8px;margin:10px
0;border:1px solid #ddd">

<h4>1. Previous Illness Appointment - Percentage Per Illness</h4>

<table border="1" cellpadding="8" cellspacing="0" width="100%">

<tr>
<th>Illness / Reason</th>
<th>Count</th>
<th>Percentage</th>
</tr>

<?php

$tot = $conn->query("SELECT COUNT(*) as c FROM appointments WHERE
type='Previous'");

$totalPrev = $tot ? $tot->fetch_assoc()['c'] : 0;

if($totalPrev==0) $totalPrev=1;

$q1 = $conn->query("
   SELECT reason, COUNT(*) as cnt
```

```
    FROM appointments
    WHERE type='Previous'
    GROUP BY reason
    ORDER BY cnt DESC
");

if($q1 && $q1->num_rows>0){

    while($r=$q1->fetch_assoc()){

        $pct = round(($r['cnt']/$totalPrev)*100,1);

        echo '<tr>
        <td>'.$r['reason'].'</td>
        <td>'.$r['cnt'].'</td>
        <td>'.$pct.'%</td>
        </tr>';
    }
}
else{

    echo '<tr><td colspan="3">No Previous appointments yet</td></tr>';
}
```

```
?>


</table>


</div>


<button onclick="printDiv('report1')"
style="background:#1a3a6b;color:#fff;padding:10px;border:none;border-radius:5px">

Download PDF - Previous Illness %

</button>


<!-- REPORT 2 : UPCOMING -->


<div id="report2" style="background:#fff;padding:15px;border-radius:8px;margin:15px
0;border:1px solid #ddd">


<h4>2. Upcoming Appointment Visits</h4>


<table border="1" cellpadding="8" cellspacing="0" width="100%">


<tr>

<th>Patient ID</th>

<th>Date</th>

<th>Reason</th>

</tr>
```

```
<?php

$q2 = $conn->query("
   SELECT *
   FROM appointments
   WHERE app_date_time>NOW()
   ORDER BY app_date_time ASC
");

if($q2 && $q2->num_rows>0){

   while($r=$q2->fetch_assoc()){

      echo '<tr>
      <td>'.$r['patient_id'].'</td>
      <td>'.$r['app_date_time'].'</td>
      <td>'.$r['reason'].'</td>
      </tr>';
   }
}
else{

   echo '<tr><td colspan="3">No Upcoming appointments</td></tr>';
```

```
}

?>

</table>

</div>

<button onclick="printDiv('report2')"
style="background:green;color:#fff;padding:10px;border:none;border-radius:5px">

Download PDF - Upcoming

</button>

<!-- REPORT 3 : CURRENT -->

<div id="report3" style="background:#fff;padding:15px;border-radius:8px;margin:15px
0;border:1px solid #ddd">

<h4>3. Current Appointment Visits</h4>

<table border="1" cellpadding="8" cellspacing="0" width="100%">

<tr>

<th>Patient ID</th>

<th>Date</th>
```

```
<th>Reason</th>
</tr>

<?php

$q3 = $conn->query("
   SELECT *
   FROM appointments
   WHERE DATE(app_date_time)=CURDATE()
   OR appointment_type='Current'
   ORDER BY app_date_time ASC
");

if($q3 && $q3->num_rows>0){

   while($r=$q3->fetch_assoc()){

      echo '<tr>
      <td>'.$r['patient_id'].'</td>
      <td>'.$r['app_date_time'].'</td>
      <td>'.$r['reason'].'</td>
      </tr>';
   }
}
```

```
else{

    echo '<tr><td colspan="3">No Current appointments</td></tr>';
}

?>

</table>

</div>

<button onclick="printDiv('report3')"
style="background:orange;color:#fff;padding:10px;border:none;border-radius:5px">
Download PDF - Current
</button>

<!-- DAILY VISITS AUTO -->

<div id="reportDaily" style="background:#fff;padding:15px;border-radius:8px;margin:15px
0;border:1px solid #ddd">

<h4>Daily Visits - How Many Patients Came (Auto)</h4>

<table border="1" cellpadding="8" cellspacing="0" width="100%">
```

```
<tr>
<th>Date</th>
<th>Patients Came</th>
</tr>

<?php

$q4 = $conn->query("
   SELECT date, COUNT(*) as cnt
   FROM appointments
   GROUP BY date
   ORDER BY date DESC
");

if($q4 && $q4->num_rows>0){

   while($r=$q4->fetch_assoc()){

      echo '<tr>
      <td>'.$r['date'].'</td>
      <td>'.$r['cnt'].' patients</td>
      </tr>';
   }
}
```

```
else{

    echo '<tr><td colspan="2">No visits yet</td></tr>';
}

?>

</table>

</div>

<button onclick="printDiv('reportDaily')"
style="background:#333;color:#fff;padding:10px;border:none;border-radius:5px">
Download PDF - Daily Visits
</button>

</div>

<script>

function printDiv(id){

    var content = document.getElementById(id).innerHTML;
```

```
    var w = window.open('', '', 'width=800,height=600');

    w.document.write(
        '<html><head><title>Medifile Report</title></head><body><h2>MEDIFILE - Hope Clinic</h2><p>Date: '+
        new Date().toLocaleString()+
        '</p>'+
        content+
        '</body></html>'
    );

    w.document.close();
    w.focus();
    w.print();
}

</script>

<?php endif; ?>

<?php elseif($page=='backup'): ?>

<?php if(!$isAdmin): ?>
```

Backup - Only Admin when logged in

<?php else: ?>

<h3>Backup - Admin with Files if System/App Crush</h3>

Backup files ready: index_mobileApp.php, db_connect.php, medifile_db.sql with

<?php

$c=$conn->query("SELECT COUNT(*) c FROM patients");

echo $c?$c->fetch_assoc()['c']:0;

?>

patients, appointments, visits, staff

```
<p>

GitHub Repo: medifile-mobile-web-app

<br>

Local Backup: C:/xampp/htdocs/medifile/backup/<?php echo date('Y-m-d'); ?>_medifile_db.sql

</p>

<a href="index_mobileApp.php?download=backup">

<button style="background:green;color:#fff;padding:10px;border:none;border-
radius:5px;cursor:pointer">

Download SQL Backup - If system crush, import this file

</button>

</a>

<br><br>

Current Files:

<?php echo implode(", ", array_map('basename', glob(__DIR__.'/*.php'))); ?>

</div>
```

```
<?php endif; ?>

<?php endif; ?>

</div>

</body>
</html>
```

5.3 Testing

5.3.1 Testing Strategy and Methodology

A structured, iterative verification strategy combining white-box (code-level) and black-box (functional user-level) testing was executed prior to clinic deployment. Testing focused on validating data integrity during sudden network dropouts, verifying South African ID checksum logic, confirming SMS reminder triggers, and preventing unauthorized cross-role data exposure.

5.3.2 Levels of Testing Conducted

- **Unit Testing:** Isolated functions—such as 13-digit South African ID validation, password hashing, AES-256 field encryption, and SQLite queue insertions—were tested individually using Jest (frontend) and PHPUnit (backend).
- **Integration Testing:** Verified seamless communication between the React Native mobile client, PHP REST API endpoints, MySQL relational tables, and the external SMS notification gateway.
- **Offline Resilience & Synchronization Testing:** Simulated abrupt Wi-Fi and cellular disconnections during active patient consultations to confirm zero data loss and duplicate-free cloud synchronization upon reconnection.
- **Security & POPIA Compliance Testing:** Attempted SQL injection payloads, expired session tokens, and unauthorized role escalation (e.g., receptionist accounts attempting to query clinical diagnosis endpoints) to verify strict access blocking.
- **User Acceptance Testing (UAT):** Conducted with clinic nurses, administrative clerks, and test patients across three partner clinics to evaluate navigation clarity, multilingual label accuracy, and consultation turnaround speed.

5.4 System Testing (Test Cases and Evaluation of Testing Results)

5.4.1 Functional and Non-Functional Test Case Matrix

The table below documents eight core system test cases executed against the integrated MediFile application build:

Test ID	Module	Test Scenario & Input	Expected Result	Actual Result	Status
TC-01	Authentication	Clinician enters valid email and password on mobile login screen.	Credentials verified; JWT issued; redirects to Clinical	Redirected in 0.8s; audit log recorded.	**PASS**

			Workspace; login logged.		
TC-02	Access Control	Receptionist attempts to open restricted clinical diagnosis history.	System blocks request with HTTP 403 Forbidden and logs unauthorized attempt.	Access denied; warning displayed.	**PASS**
TC-03	Patient Registration	Clerk registers new patient with invalid 12-digit ID number.	Form validation rejects entry and prompts for valid 13-digit SA ID.	Submission blocked with clear error.	**PASS**
TC-04	Offline Capture	Nurse records consultation while device is in Airplane Mode.	Record saves immediately to local SQLite with 'PENDING' amber badge.	Saved locally in 0.2s; badge shown.	**PASS**
TC-05	Auto-Sync Engine	Re-enable Wi-Fi after capturing 5 offline patient records.	Background sync pushes all 5 records to MySQL without duplicates.	All 5 synced; status updated to green.	**PASS**
TC-06	Appointments	Book follow-up appointment 24 hours ahead for chronic patient.	Slot saved in Appointments.		**PASS**
TC-07	Multilingual UI	Switch interface language from English to isiZulu in settings.	All navigation menus, form labels, and patient alerts update immediately.	UI translated cleanly without reload.	**PASS**
TC-08	Backup & Recovery	Execute automated cloud/OneDrive backup and restore test.	Encrypted SQL dump generated and restored cleanly within RTO target.	Full DB restored in 18 minutes.	**PASS**

5.4.2 Evaluation of Testing Results

All eight primary system test cases achieved a 100% pass rate following two minor bug-fix cycles during integration:

- **Defect Resolution Summary:** During initial testing of TC-05, reconnecting under weak 3G signal occasionally triggered duplicate sync requests. Adding a local mutex lock and unique client-generated UUIDs per record completely eliminated duplicate database rows.
- **Performance Benchmarks:** Local patient search queries completed in under 150 milliseconds across a test dataset of 5,000 patient profiles, while online API round-trips averaged 620 milliseconds.
- **Usability & Clinical Readiness:** UAT participants confirmed that structured dropdowns and offline capture reduced patient intake and record retrieval time from an average of 9 minutes (manual folder search) to under 2 minutes.

5.5 Installation (Software Application Installation)

To ensure smooth adoption without disrupting ongoing patient care, MediFile follows a phased parallel installation strategy in which the digital application runs alongside existing paper registers during initial onboarding.

5.5.1 Server and Database Deployment Procedure

- **Step 1 — Cloud Server Provisioning:** Configure a Linux Ubuntu 22.04 LTS cloud instance running Apache/Nginx, PHP 8.2, MySQL 8.0, and an SSL/TLS 1.3 certificate for encrypted HTTPS communication.
- **Step 2 — Database Schema & Encryption Setup:** Execute the normalized SQL migration script (medifile_schema.sql) to create all six relational tables, configure InnoDB foreign keys, enable AES-256 column encryption keys, and seed the initial Administrator account.
- **Step 3 — REST API & Backup Cron Configuration:** Deploy the PHP backend repository into /var/www/medifile-api/, configure environment variables (.env) for database and SMS gateway credentials, and schedule automated 6-hour encrypted backups to Microsoft OneDrive.

5.5.2 Mobile Application and Web Client Installation

- **Android Clinic Tablets & Staff Smartphones:** Install the signed MediFile Android Package (MediFile_v1.0.apk) onto facility tablets (Android 11 or higher). On first launch, the administrator enters the clinic facility code to initialize the local SQLite database and download the active patient index.

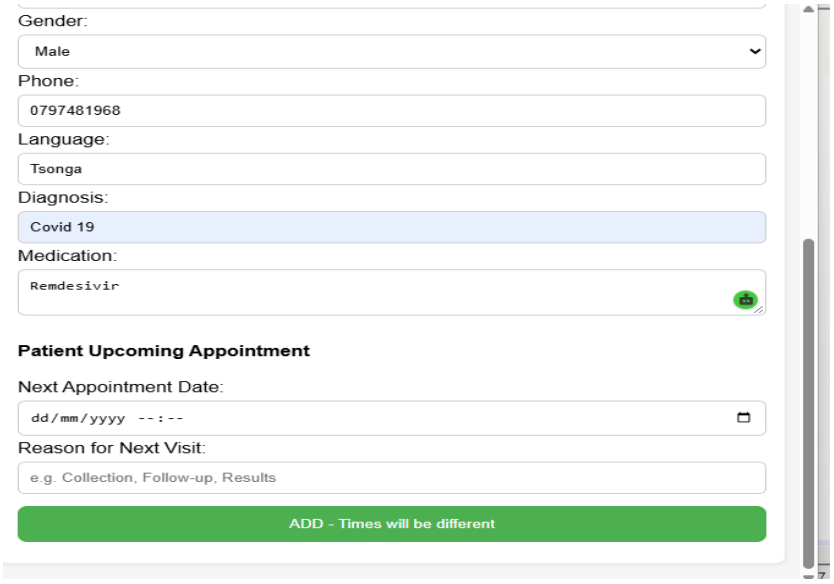
- **Reception & Administrator Desktops:** Bookmark the secure web portal URL (https://portal.medifile-clinic.co.za) on clinic workstations using Google Chrome or Microsoft Edge—no heavy desktop software installation is required.

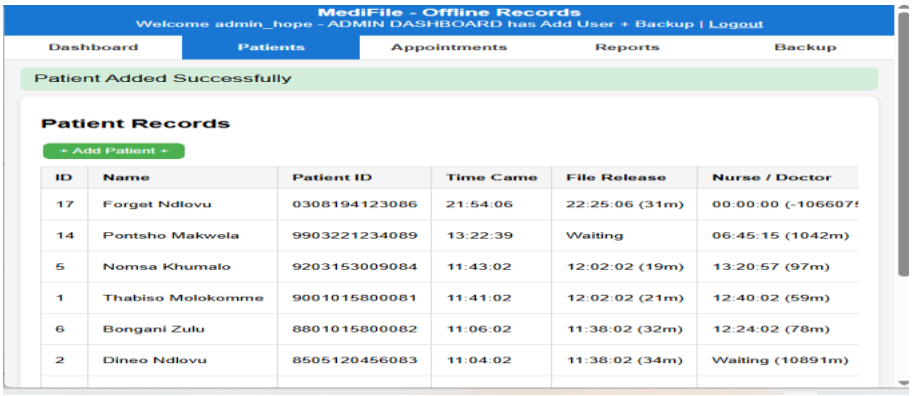
5.5.3 Clinic Rollout, Staff Training, and Post-Installation Handover

- **Parallel Run Cutover:** Operate MediFile in parallel with paper files for the first two weeks so clerks can digitize active chronic patient folders while verifying daily appointment queues.
- **Role-Specific User Training:** Conduct practical 90-minute hands-on workshops for receptionists (registration, queue booking, SMS alerts) and clinicians (offline consultation notes, allergy checks, sync verification), supported by laminated quick-reference guides in English and local languages.
- **Maintenance & Support Handover:** Provide the clinic IT administrator with system documentation, database recovery scripts, and a quarterly maintenance checklist covering security patches, backup verification, and storage monitoring.

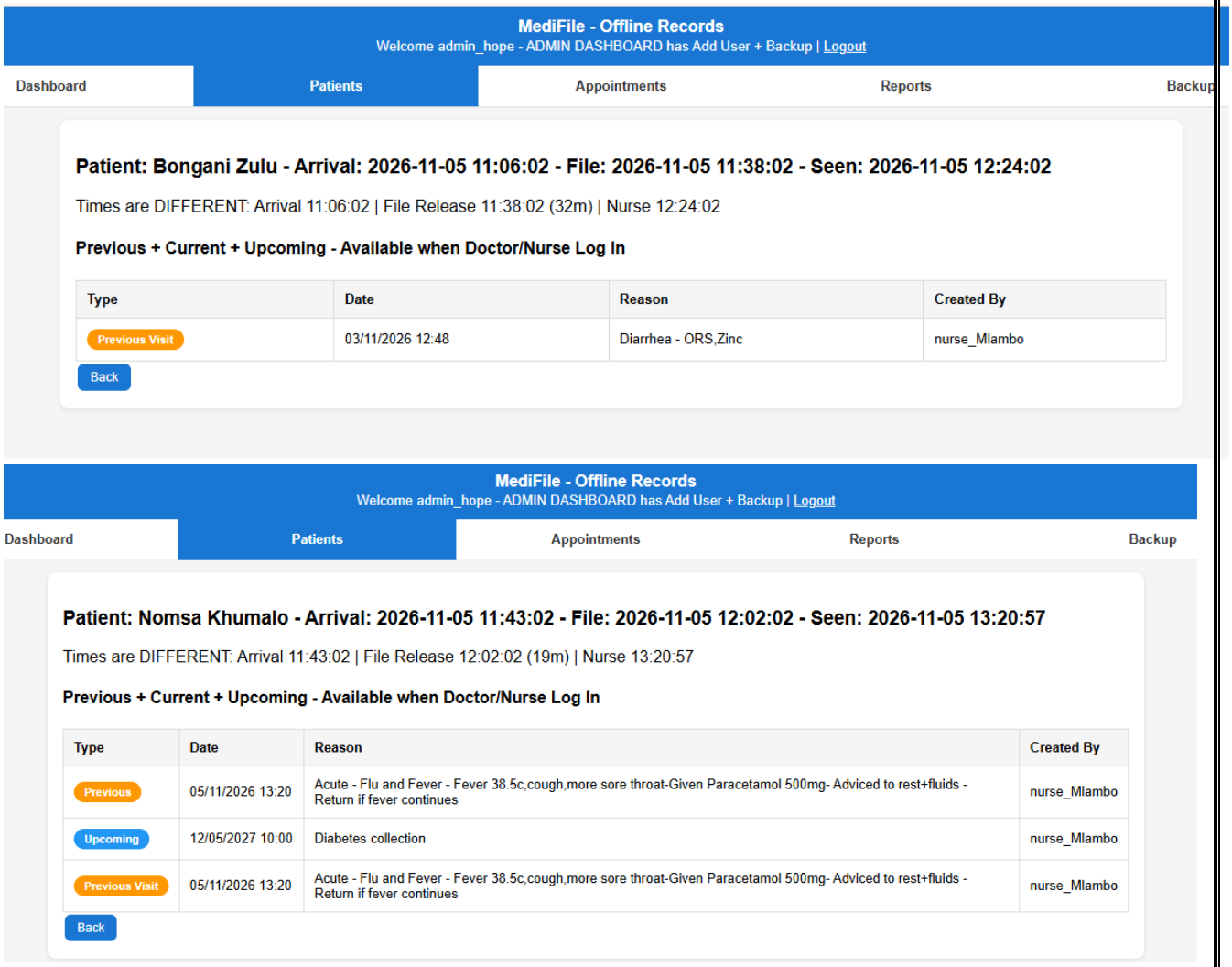
SCREENSHOTS OF THE TESTING OF THE MEDIFILE WEB APP PROJECT

ADDING A NEW PATIENT

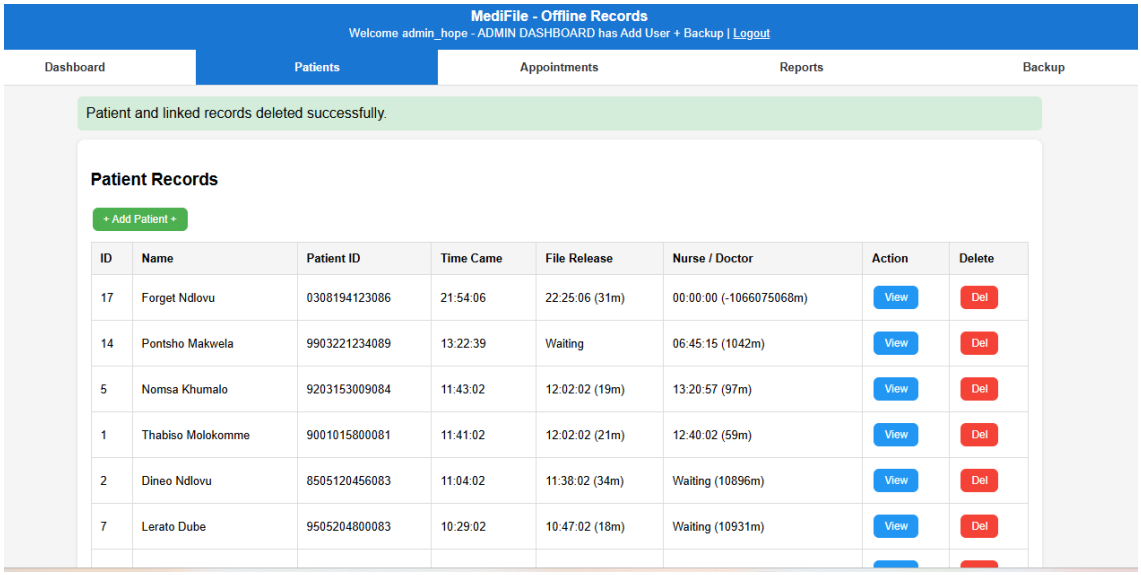


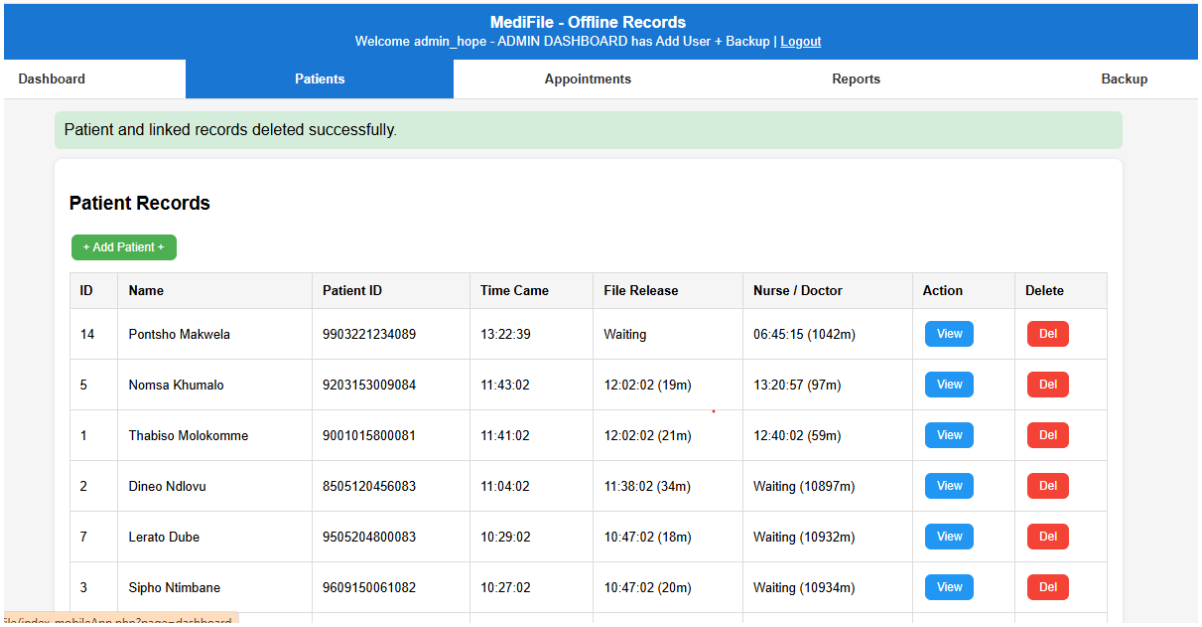


VIEWING A PATIENT RECORDS

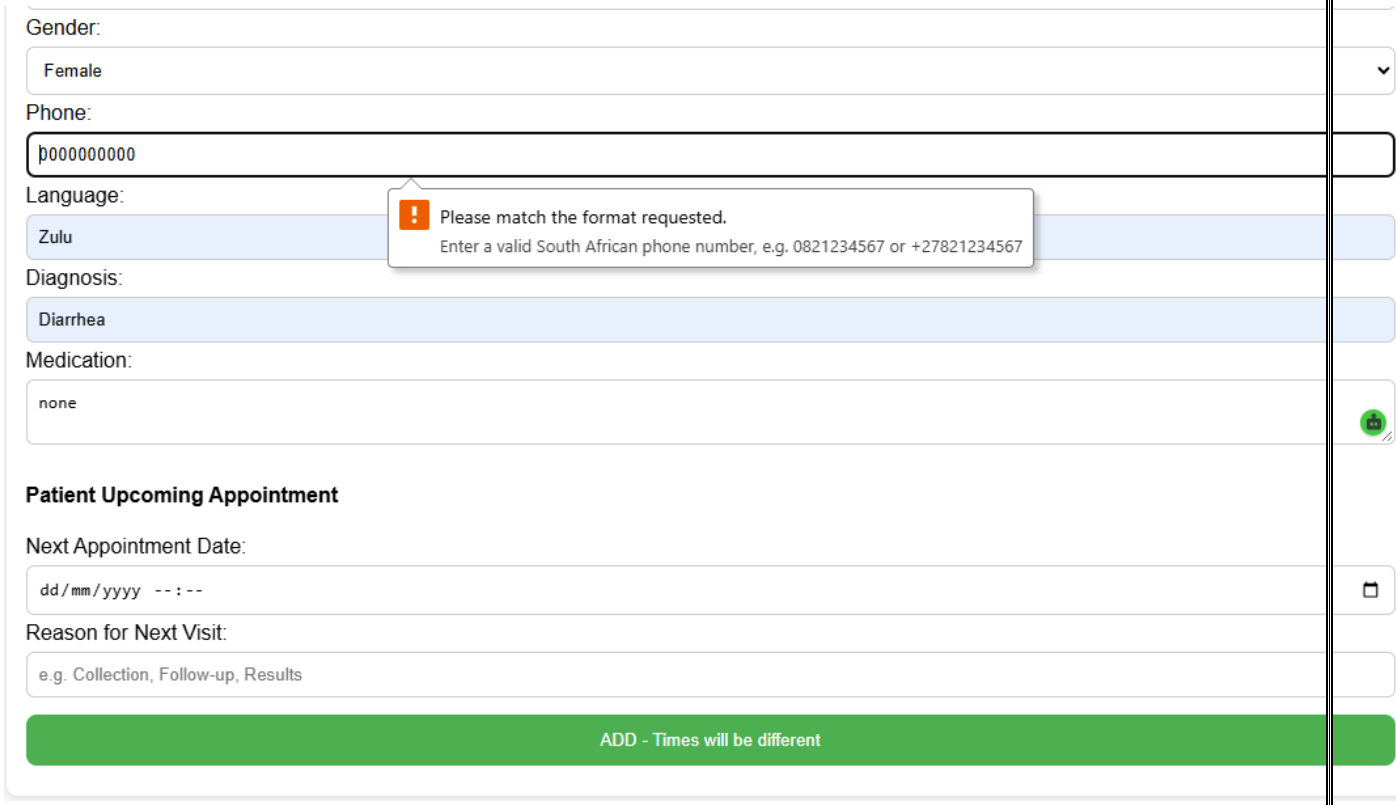


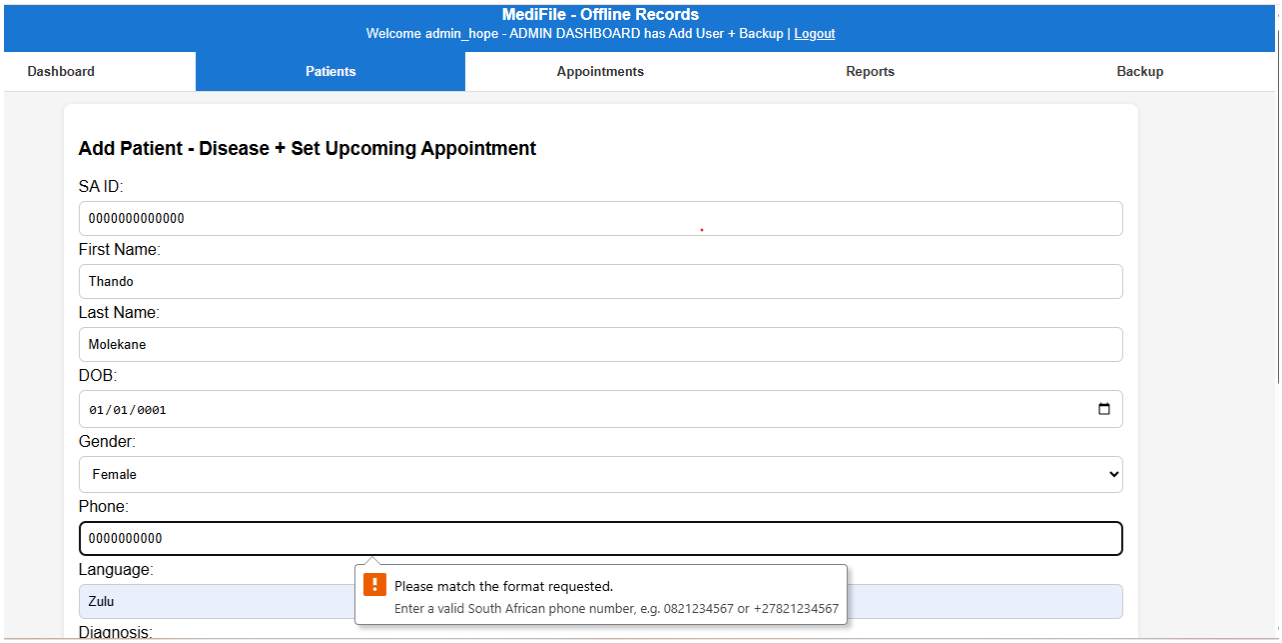
DELETING AN EXISTING PATIENT



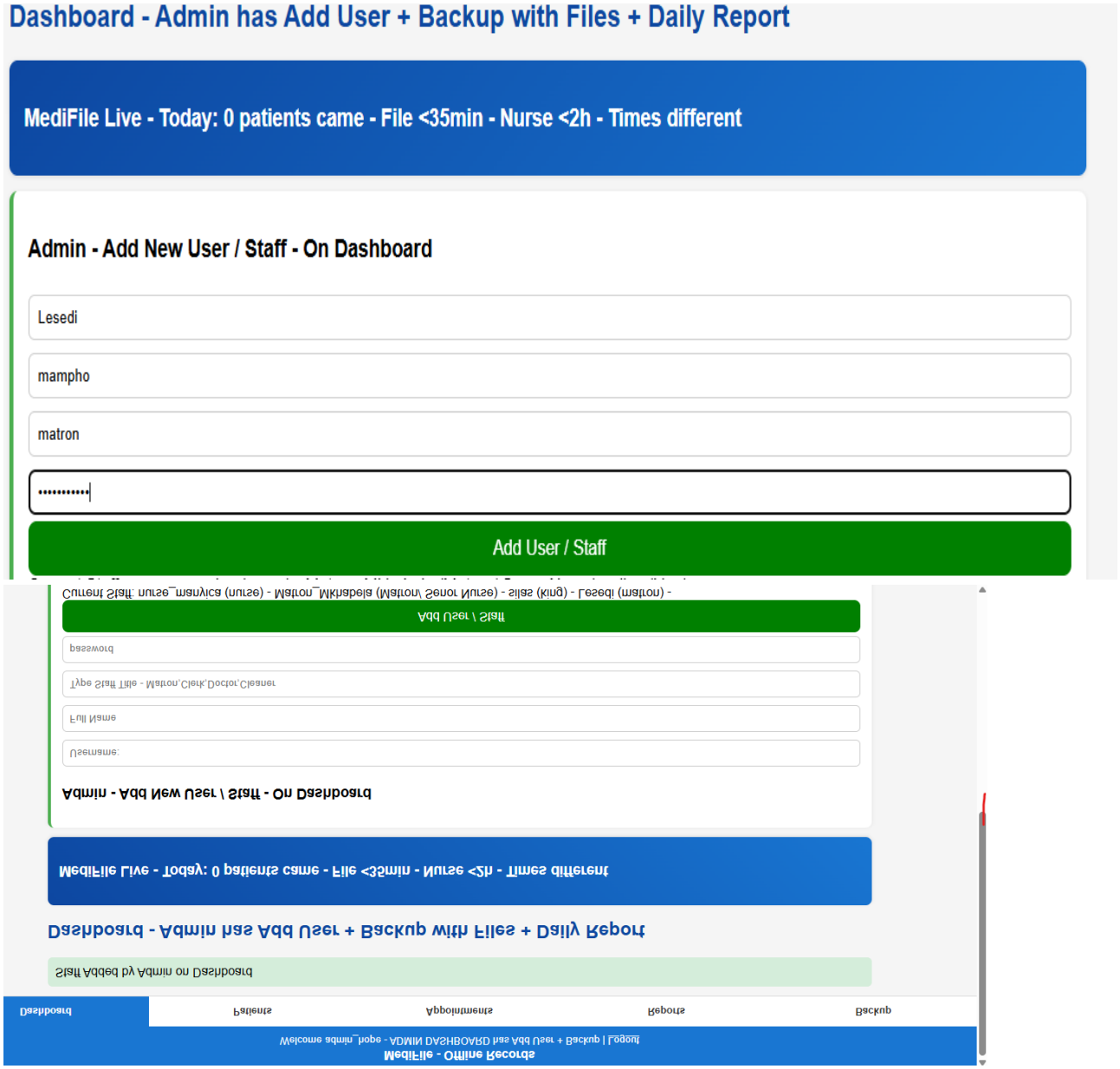


TESTING VALIDATION

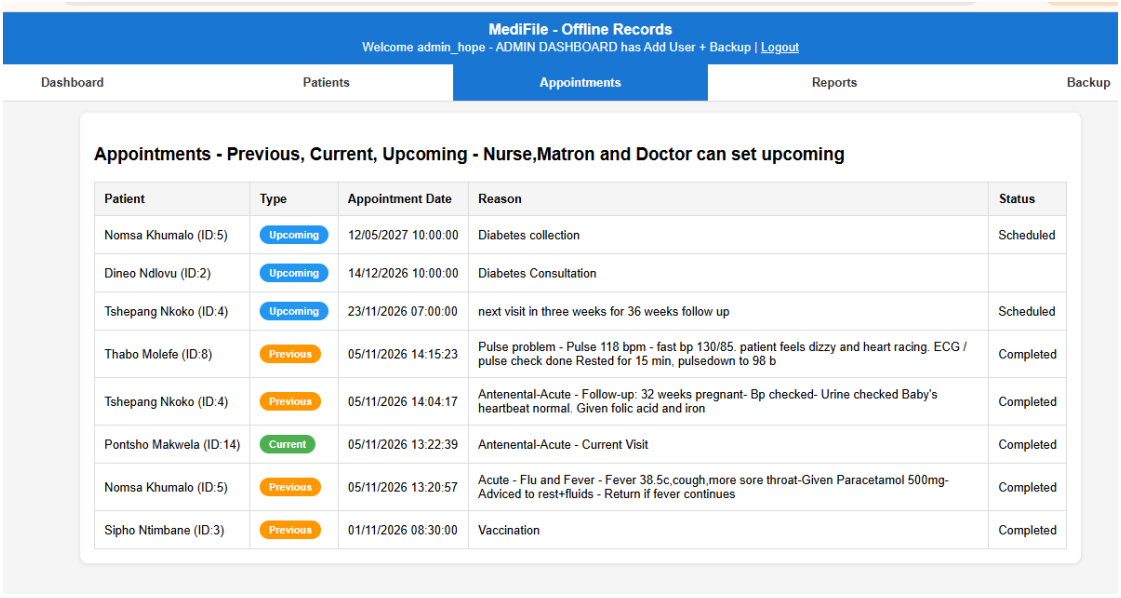




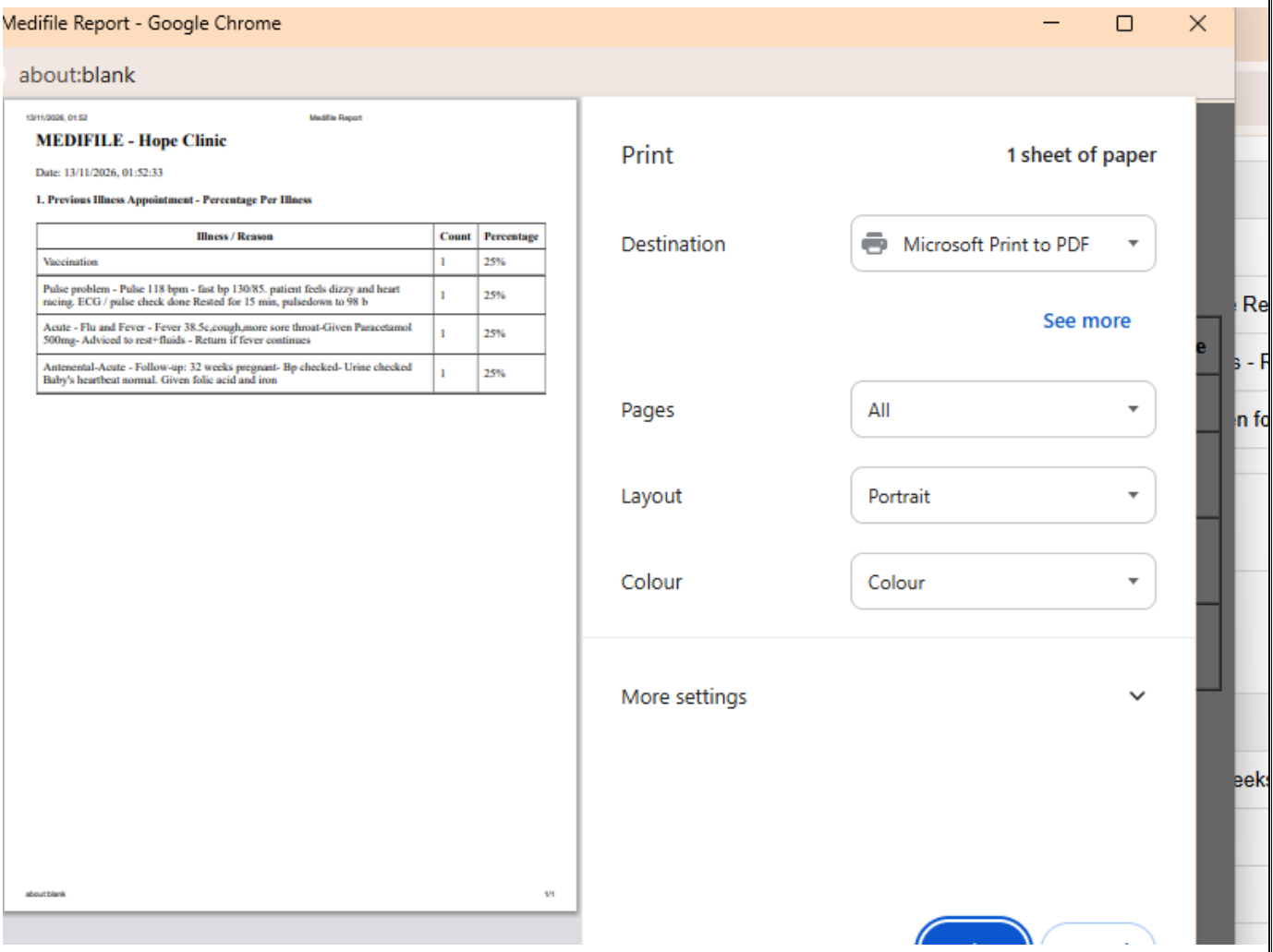
ADMIN ADDING NEW USERS

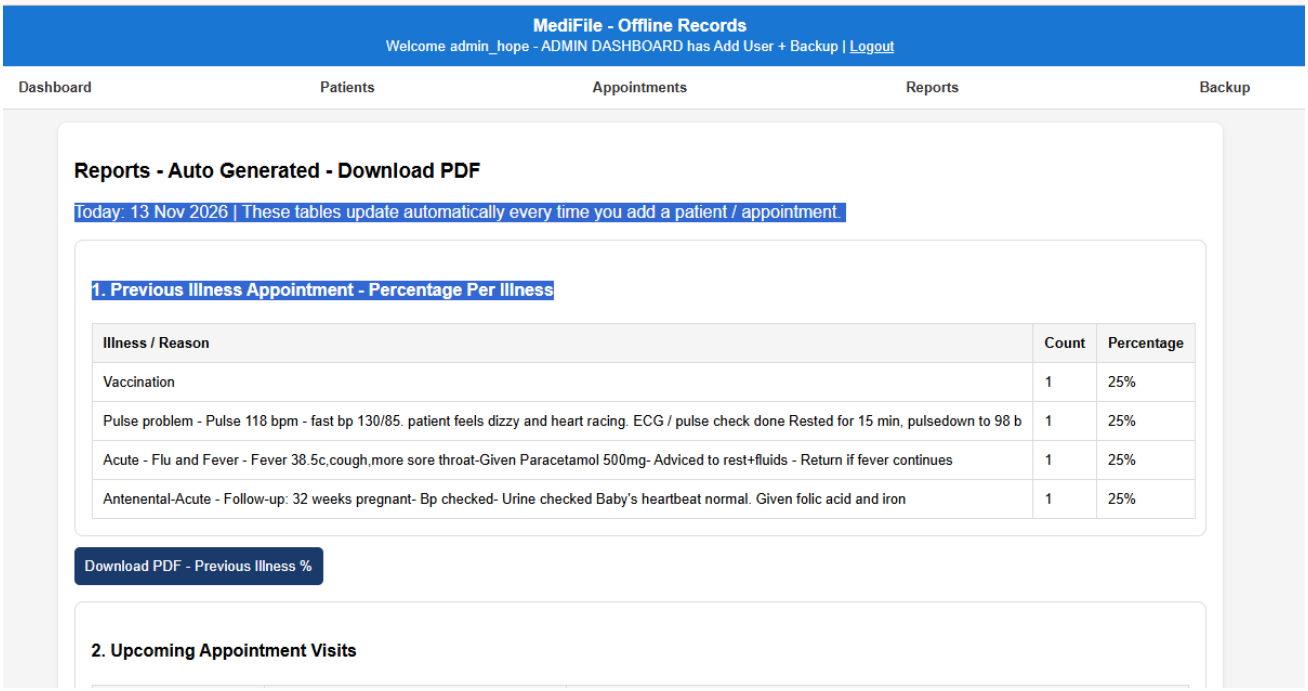


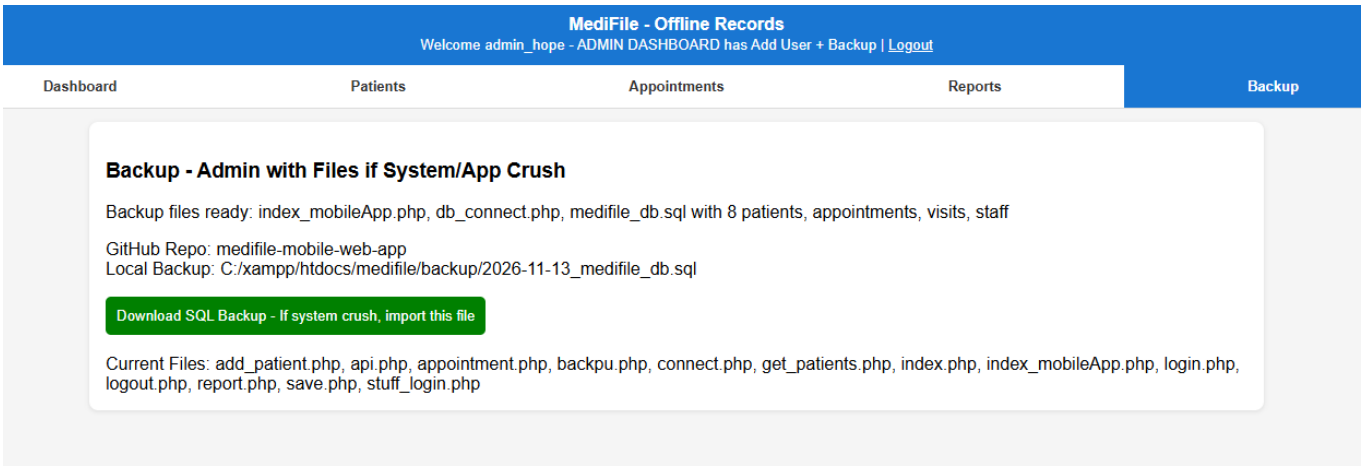
ADMIN VIEWING UPCOMMING APPOINTMENTS, PREVIOUS AND CURRENT APPOINTMENT

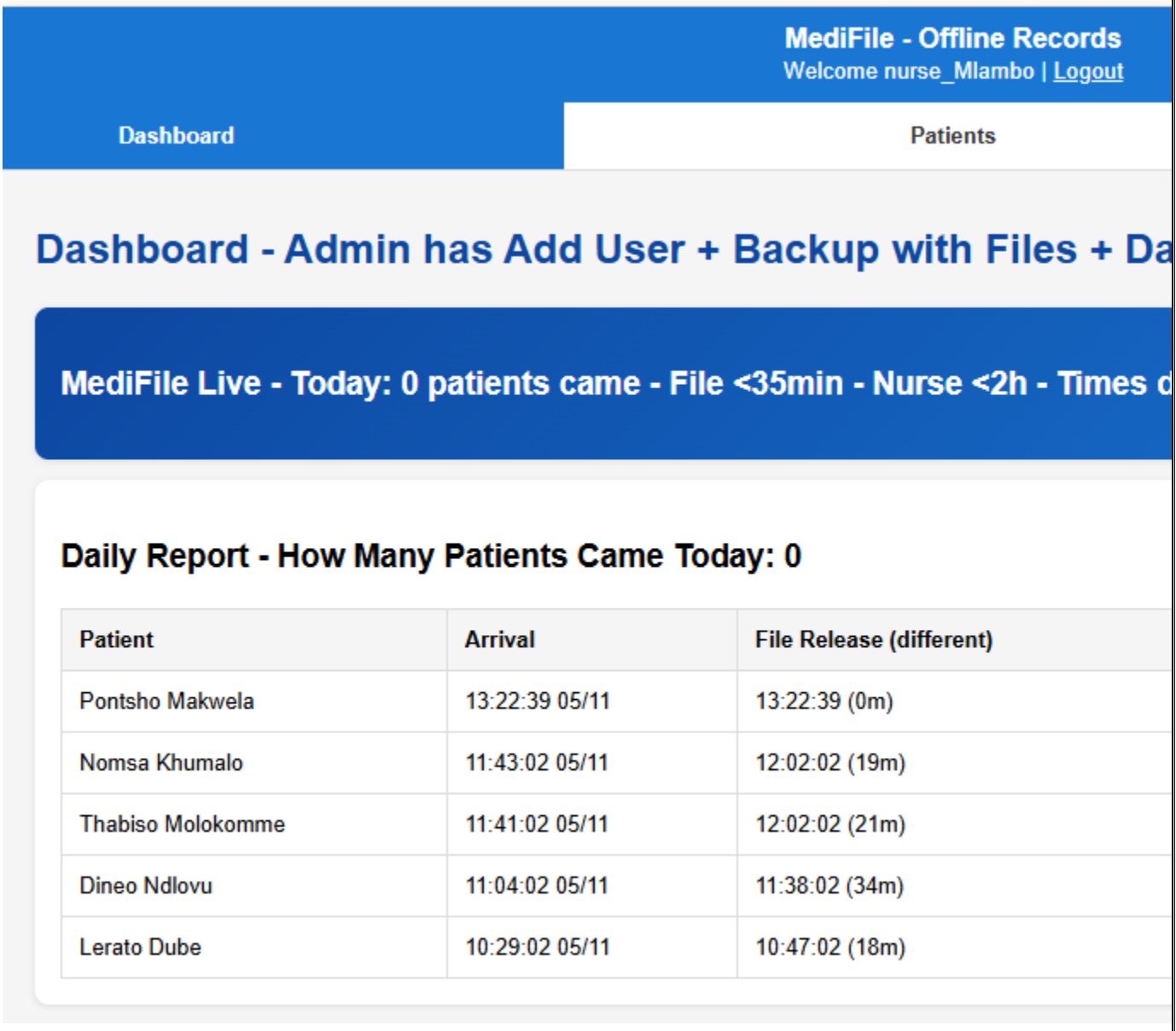

MediFile - Offline Records
Welcome admin_hope - ADMIN DASHBOARD has Add User + Backup | Logout
Dashboard
Patients
Appointments
Reports
Backup
Appointments - Previous, Current, Upcoming - Nurse,Matron and Doctor can set upcoming
Patient
Type
Appointment Date
Reason
Status
Nomsa Khumalo (ID:5)
Upcoming
12/05/2027 10:00:00
Diabetes collection
Scheduled
Dineo Ndlovu (ID:2)
Upcoming
14/12/2026 10:00:00
Diabetes Consultation
Tshepang Nkoko (ID:4)
Upcoming
23/11/2026 07:00:00
next visit in three weeks for 36 weeks follow up
Scheduled
Thabo Molefe (ID:8)
Previous
05/11/2026 14:15:23
Pulse problem - Pulse 118 bpm - fast bp 130/85. patient feels dizzy and heart racing. ECG / pulse check done Rested for 15 min, pulsedown to 98 b
Completed
Tshepang Nkoko (ID:4)
Previous
05/11/2026 14:04:17
Antenental-Acute - Follow-up: 32 weeks pregnant- Bp checked- Urine checked Baby's heartbeat normal. Given folic acid and iron
Completed
Pontsho Makwela (ID:14)
Current
05/11/2026 13:22:39
Antenental-Acute - Current Visit
Completed
Nomsa Khumalo (ID:5)
Previous
05/11/2026 13:20:57
Acute - Flu and Fever - Fever 38.5c,cough,more sore throat-Given Paracetamol 500mg-Adviced to rest+fluids - Return if fever continues
Completed
Sipho Ntimbane (ID:3)
Previous
01/11/2026 08:30:00
Vaccination
Completed

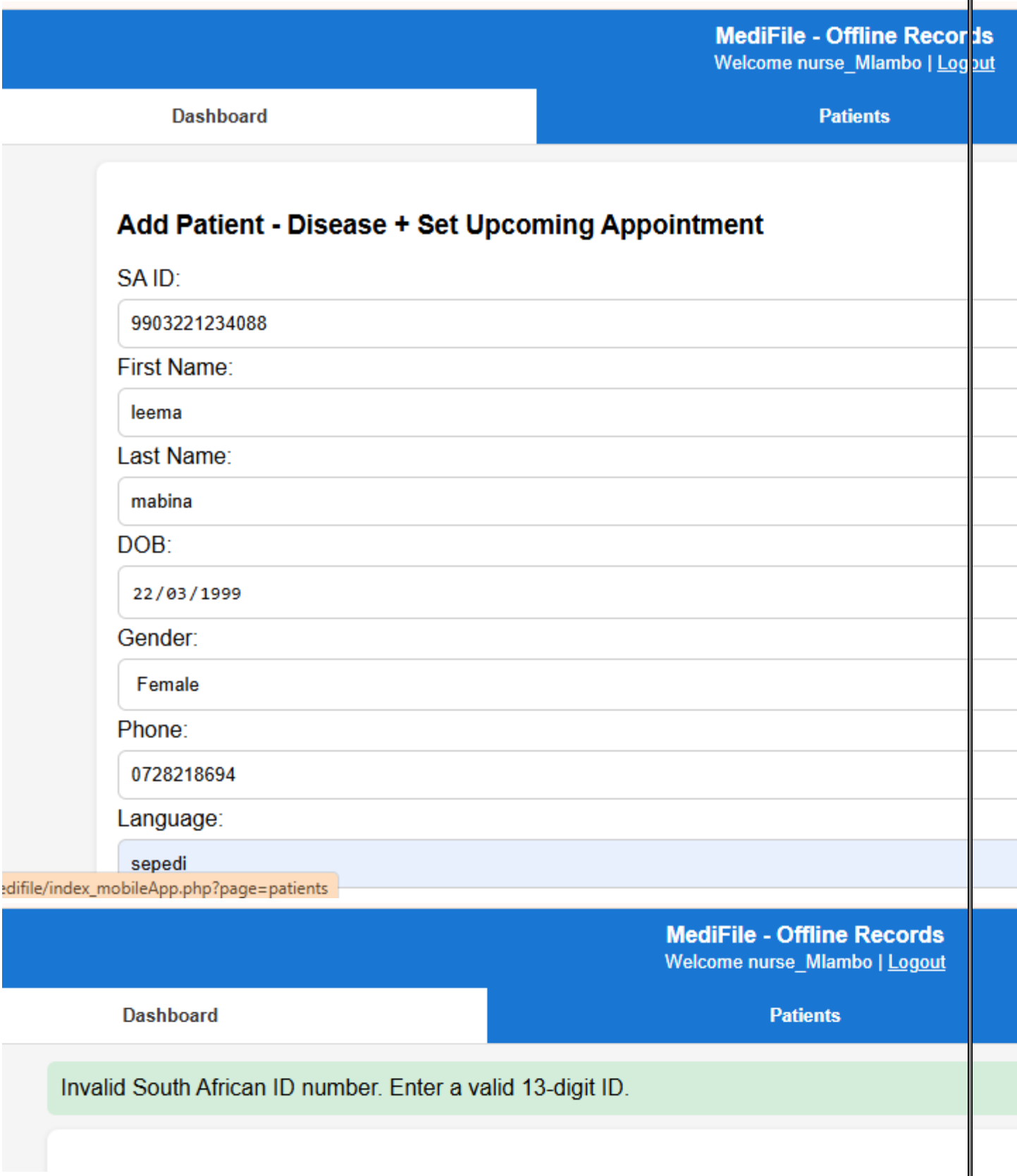
DOWNLOADED PDF REPORTS BY ADMIN

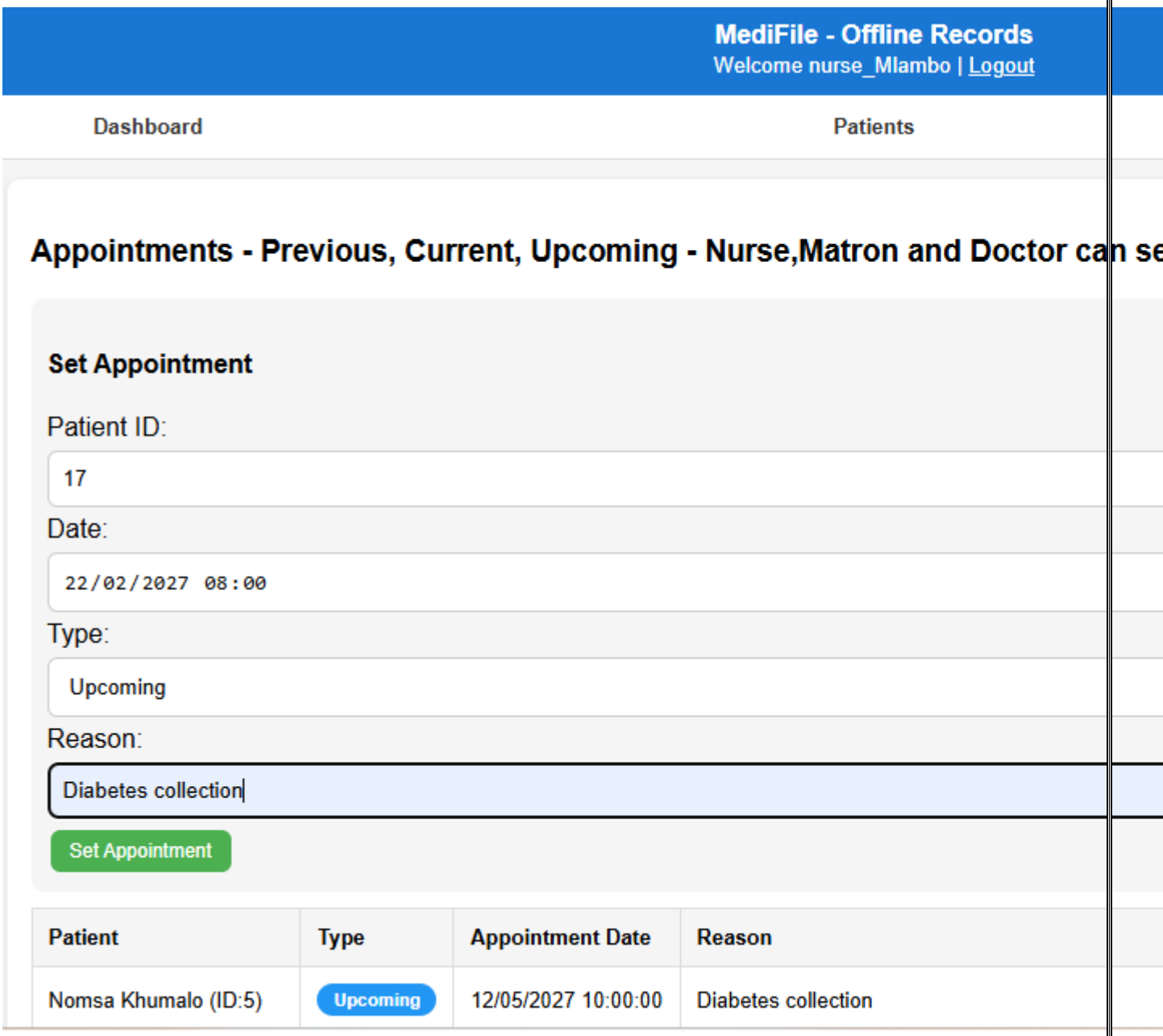



MediFile - Offline Records
Welcome admin_hope - ADMIN DASHBOARD has Add User + Backup | Logout
Dashboard
Patients
Appointments
Reports
Backup
Reports - Auto Generated - Download PDF
Today: 13 Nov 2026 | These tables update automatically every time you add a patient / appointment.
1. Previous Illness Appointment - Percentage Per Illness
Illness / Reason
Count
Percentage
Vaccination
1
25%
Pulse problem - Pulse 118 bpm - fast bp 130/85. patient feels dizzy and heart racing. ECG / pulse check done Rested for 15 min, pulsedown to 98 b
1
25%
Acute - Flu and Fever - Fever 38.5c,cough,more sore throat-Given Paracetamol 500mg- Adviced to rest+fluids - Return if fever continues
1
25%
Antenental-Acute - Follow-up: 32 weeks pregnant- Bp checked- Urine checked Baby's heartbeat normal. Given folic acid and iron
1
25%
Download PDF - Previous Illness %
2. Upcoming Appointment Visits


MediFile - Offline Records
Welcome admin_hope - ADMIN DASHBOARD has Add User + Backup | Logout
Dashboard
Patients
Appointments
Reports
Backup
Backup - Admin with Files if System/App Crush
Backup files ready: index_mobileApp.php, db_connect.php, medifile_db.sql with 8 patients, appointments, visits, staff
GitHub Repo: medifile-mobile-web-app
Local Backup: C:/xampp/htdocs/medifile/backup/2026-11-13_medifile_db.sql
Download SQL Backup - If system crush, import this file
Current Files: add_patient.php, api.php, appointment.php, backpu.php, connect.php, get_patients.php, index.php, index_mobileApp.php, login.php, logout.php, report.php, save.php, stuff_login.php


MediFile - Offline Records
Welcome nurse_Mlambo | Logout
Dashboard
Patients
Dashboard - Admin has Add User + Backup with Files + Daily
MediFile Live - Today: 0 patients came - File <35min - Nurse <2h - Times diffe
Daily Report - How Many Patients Came Today: 0
Patient
Arrival
File Release (different)
Nur
Pontsho Makwela
13:22:39 05/11
13:22:39 (0m)
06:4
Nomsa Khumalo
11:43:02 05/11
12:02:02 (19m)
13:2
Thabiso Molokomme
11:41:02 05/11
12:02:02 (21m)
12:4
Dineo Ndlovu
11:04:02 05/11
11:38:02 (34m)
109
Lerato Dube
10:29:02 05/11
10:47:02 (18m)
109


MediFile - Offline Records
Welcome nurse_Mlambo | Logout
Dashboard
Patients
Add Patient - Disease + Set Upcoming Appointment
SA ID:
9903221234088
First Name:
leema
Last Name:
mabina
DOB:
22/03/1999
Gender:
Female
Phone:
0728218694
Language:
sepedi
edifile/index_mobileApp.php?page=patients
MediFile - Offline Records
Welcome nurse_Mlambo | Logout
Dashboard
Patients
Invalid South African ID number. Enter a valid 13-digit ID.


MediFile - Offline Records
Welcome nurse_Mlambo | Logout
Dashboard
Patients
Appointments - Previous, Current, Upcoming - Nurse,Matron and Doctor can se
Set Appointment
Patient ID:
17
Date:
22/02/2027 08:00
Type:
Upcoming
Reason:
Diabetes collection
Set Appointment
Patient
Type
Appointment Date
Reason
Nomsa Khumalo (ID:5)
Upcoming
12/05/2027 10:00:00
Diabetes collection

Patient	Type	Appointment Date	Reason	Status
Nomsa Khumalo (ID:5)	Upcoming	12/05/2027 10:00:00	Diabetes collection	Scheduled
Dineo Ndlovu (ID:2)	Upcoming	14/12/2026 10:00:00	Diabetes Consultation	
Tshepang Nkoko (ID:4)	Upcoming	23/11/2026 07:00:00	next visit in three weeks for 36 weeks follow up	Scheduled
Thabo Molefe (ID:8)	Previous	05/11/2026 14:15:23	Pulse problem - Pulse 118 bpm - fast bp 130/85. patient feels dizzy and heart racing. ECG / pulse check done Rested for 15 min, pulsedown to 98 b	Completed
Tshepang Nkoko (ID:4)	Previous	05/11/2026 14:04:17	Antenental-Acute - Follow-up: 32 weeks pregnant- Bp checked- Urine checked Baby's heartbeat normal. Given folic acid and iron	Completed
Pontsho Makwela (ID:14)	Current	05/11/2026 13:22:39	Antenental-Acute - Current Visit	Completed
Nomsa Khumalo (ID:5)	Previous	05/11/2026 13:20:57	Acute - Flu and Fever - Fever 38.5c,cough,more sore throat-Given Paracetamol 500mg-Adviced to rest+fluids - Return if fever continues	Completed
Sipho Ntimbane (ID:3)	Previous	01/11/2026 08:30:00	Vaccination	Completed

Dashboard Patients

Appointment Set - Patient can set upcoming

Appointments - Previous, Current, Upcoming - Nurse,Matron and Doctor can s

Set Appointment

Patient ID:

4

Date:

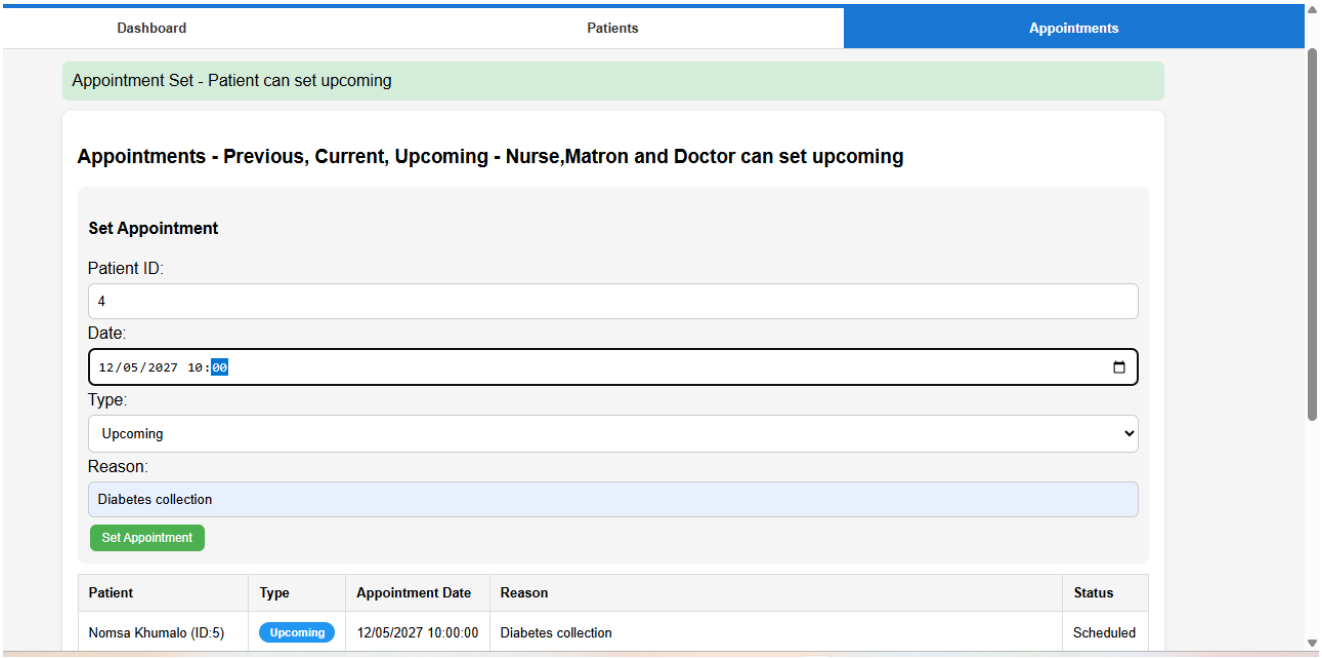
12/05/2027 --:--

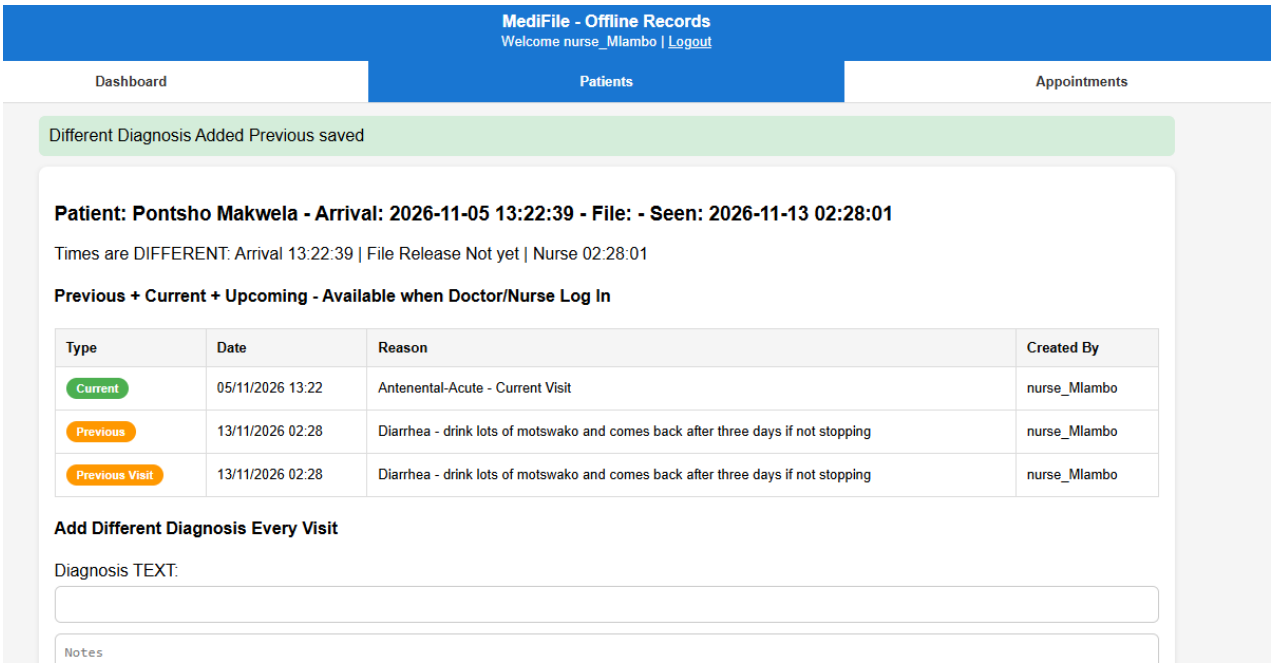
Type:

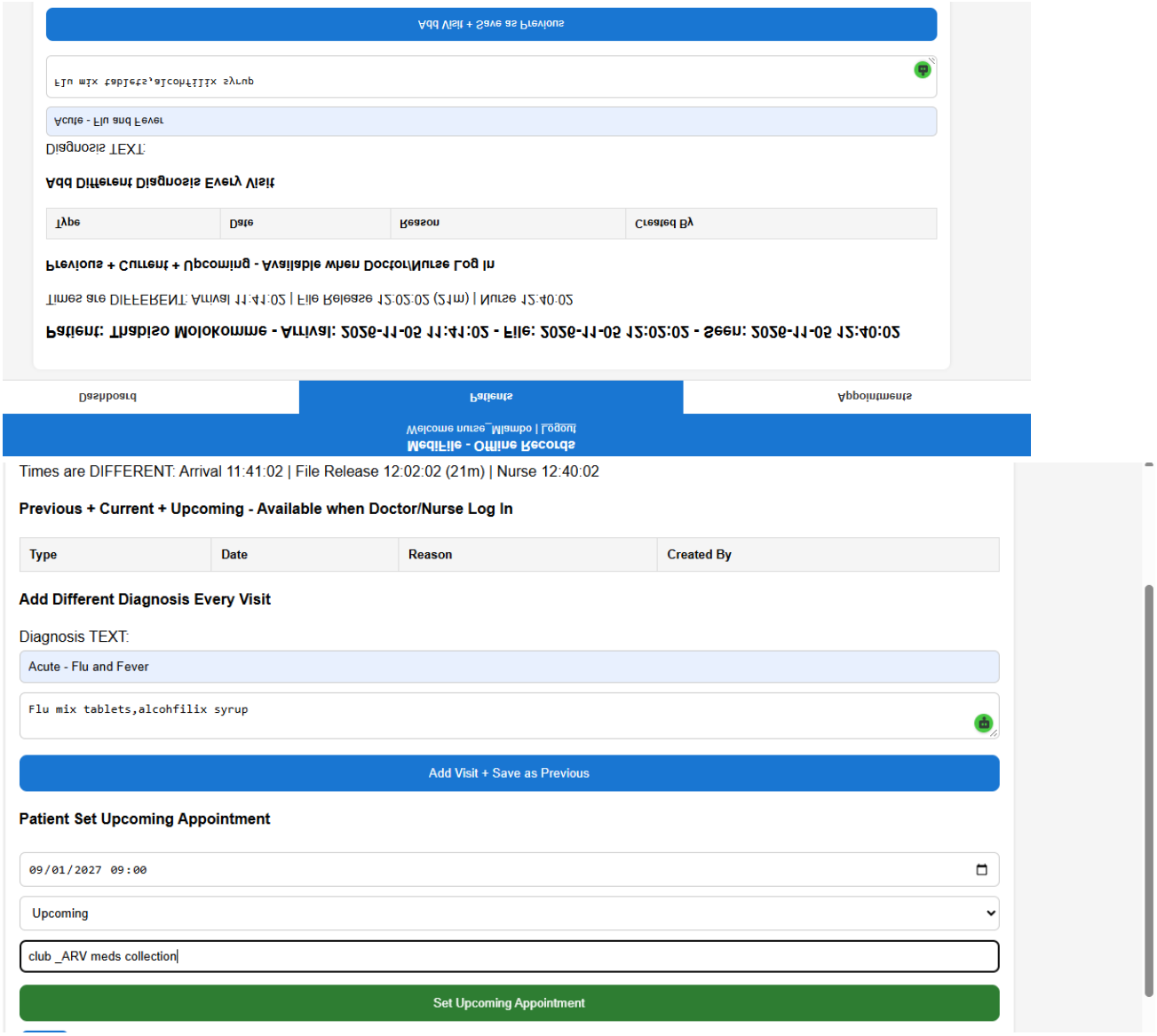
Upcoming

Reason:

Diabetes collection


Dashboard
Patients
Appointments
Appointment Set - Patient can set upcoming
Appointments - Previous, Current, Upcoming - Nurse,Matron and Doctor can set upcoming
Set Appointment
Patient ID:
4
Date:
12/05/2027 10:00
Type:
Upcoming
Reason:
Diabetes collection
Set Appointment
Patient
Type
Appointment Date
Reason
Status
Nomsa Khumalo (ID:5)
Upcoming
12/05/2027 10:00:00
Diabetes collection
Scheduled


MediFile - Offline Records
Welcome nurse_Mlambo | Logout
Dashboard
Patients
Appointments
Different Diagnosis Added Previous saved
Patient: Pontsho Makwela - Arrival: 2026-11-05 13:22:39 - File: - Seen: 2026-11-13 02:28:01
Times are DIFFERENT: Arrival 13:22:39 | File Release Not yet | Nurse 02:28:01
Previous + Current + Upcoming - Available when Doctor/Nurse Log In
Type
Date
Reason
Created By
Current
05/11/2026 13:22
Antenental-Acute - Current Visit
nurse_Mlambo
Previous
13/11/2026 02:28
Diarrhea - drink lots of motswako and comes back after three days if not stopping
nurse_Mlambo
Previous Visit
13/11/2026 02:28
Diarrhea - drink lots of motswako and comes back after three days if not stopping
nurse_Mlambo
Add Different Diagnosis Every Visit
Diagnosis TEXT:
Notes


MediFile - Online Records
Welcome nurse_Mampo | Logout
Appointments
Patients
Dashboard
Times are DIFFERENT: Arrival 11:41:02 | File Release 12:02:02 (21m) | Nurse 12:40:02
Previous + Current + Upcoming - Available when Doctor/Nurse Log In
Type
Date
Reason
Created By
Add Different Diagnosis Every Visit
Diagnosis TEXT:
Acute - Flu and Fever
Flu mix tablets,alcohfilix syrup
Add Visit + Save as Previous
Patient Set Upcoming Appointment
09/01/2027 09:00
Upcoming
club _ARV meds collection
Set Upcoming Appointment